ggplot2

Visualizando conjuntos de dados

James Sampaio, PhD

ggplot2

Visualizando conjuntos de dados

- Fundamentos
- Dispersão
- Mapeamento estético (aesthetics)
- Subdivisões gráficas (faceting)
- Suavização
- Boxplot

O gráfico simples trouxe mais informações à mente do analista de dados do que qualquer outro dispositivo.

- John Tukey

iris

```
> str(iris)
'data.frame':   150 obs. of  5 variables:
 $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
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 $ Species     : Factor w/ 3 levels "setosa","versicolor",..: 1 1 1 1 1 1 1 1 1 1 ...
```

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Rows: 150
Columns: 5
$ Sepal.Length <dbl> 5.1, 4.9, 4.7, 4.6, 5.0, 5.4, 4.6, 5.0, 4.4, 4.9, 5.4, 4.8, 4.8,…
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Comprimento da sépala

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Largura da sépala

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Comprimento da pétala

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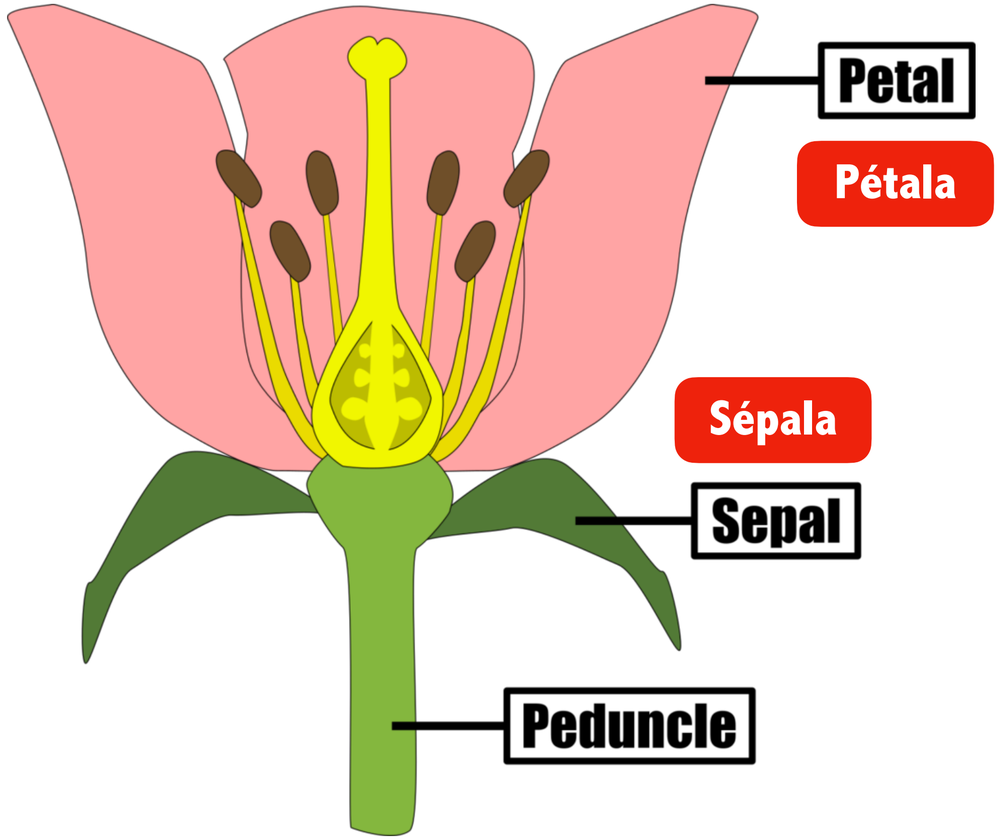
```

Espécie

iris

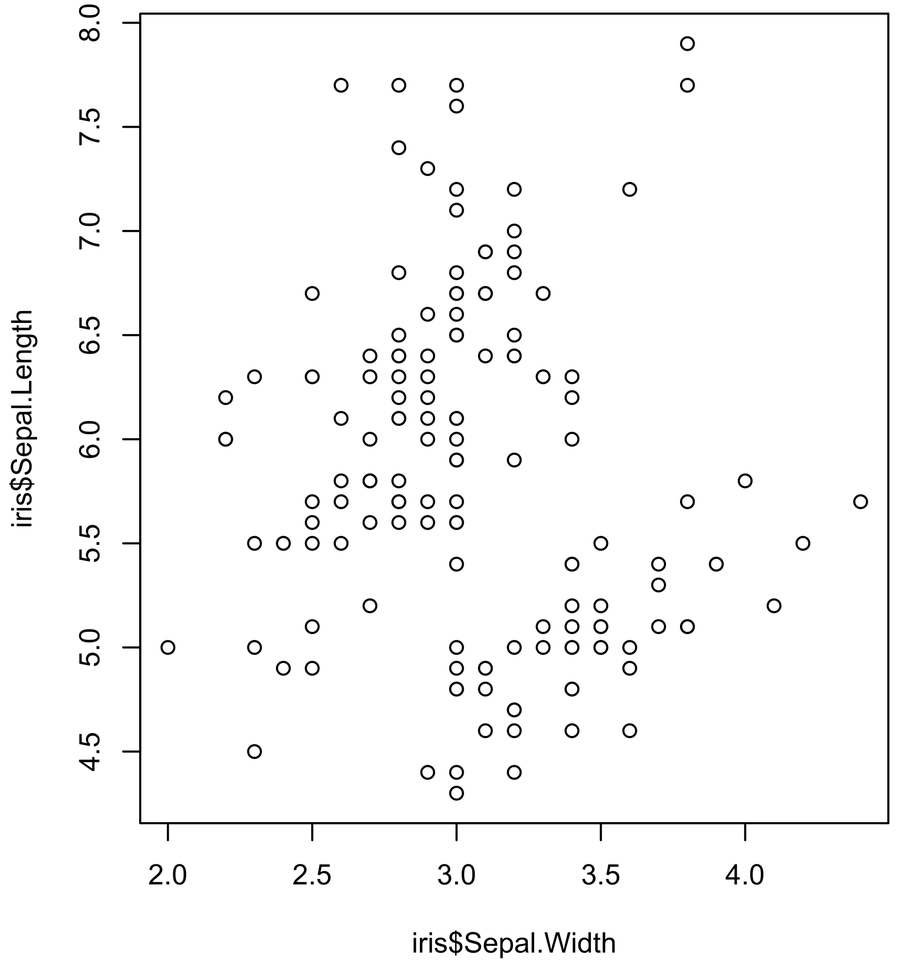


Plot

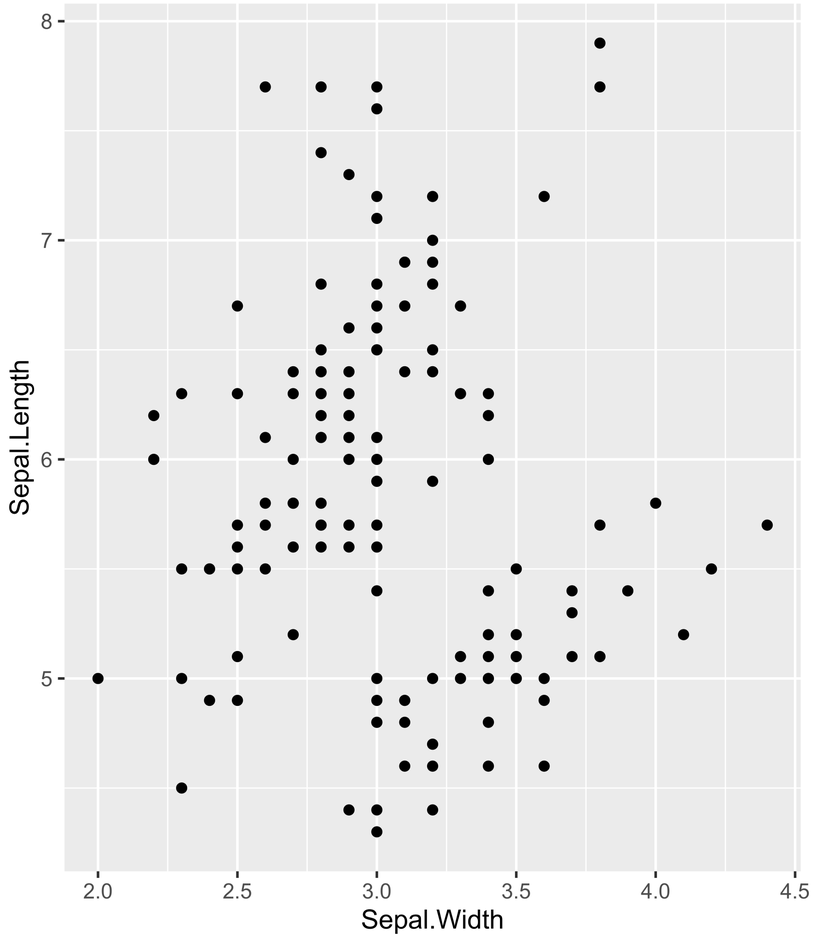
R base

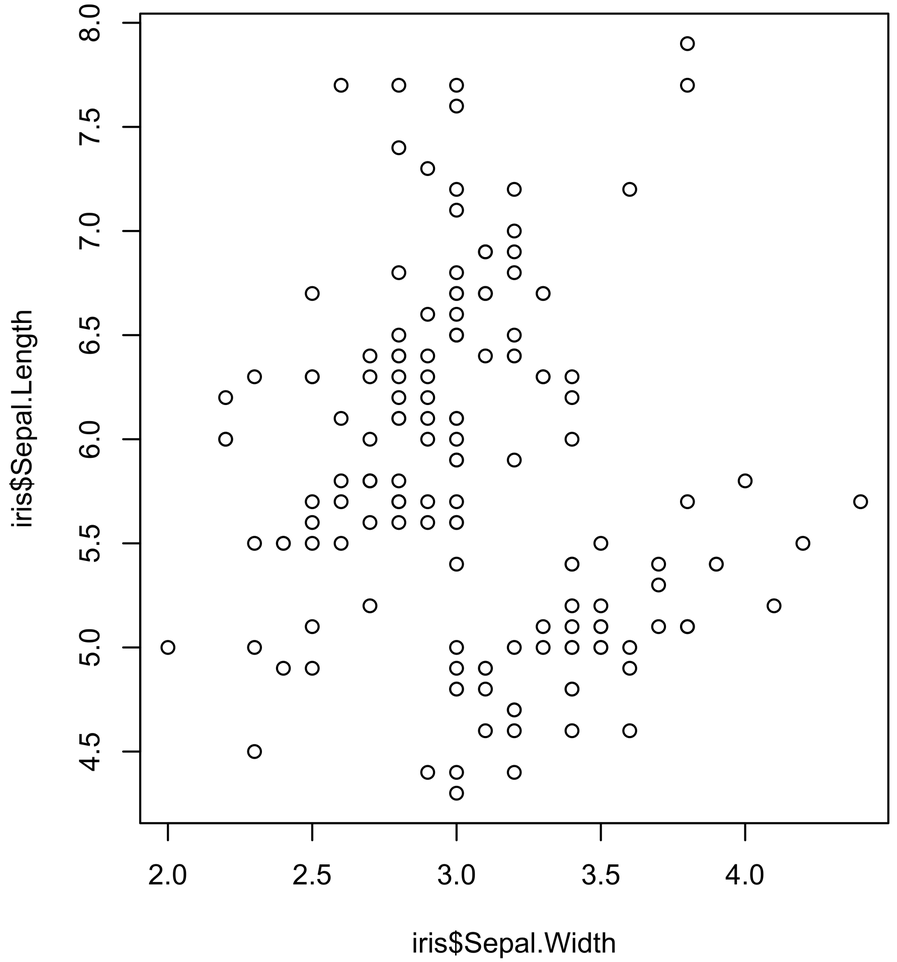
```
plot(iris$Sepal.Width,
     iris$Sepal.Length)
```

- Método básico do R para gráficos
- Simples
- Difícil de personalizar

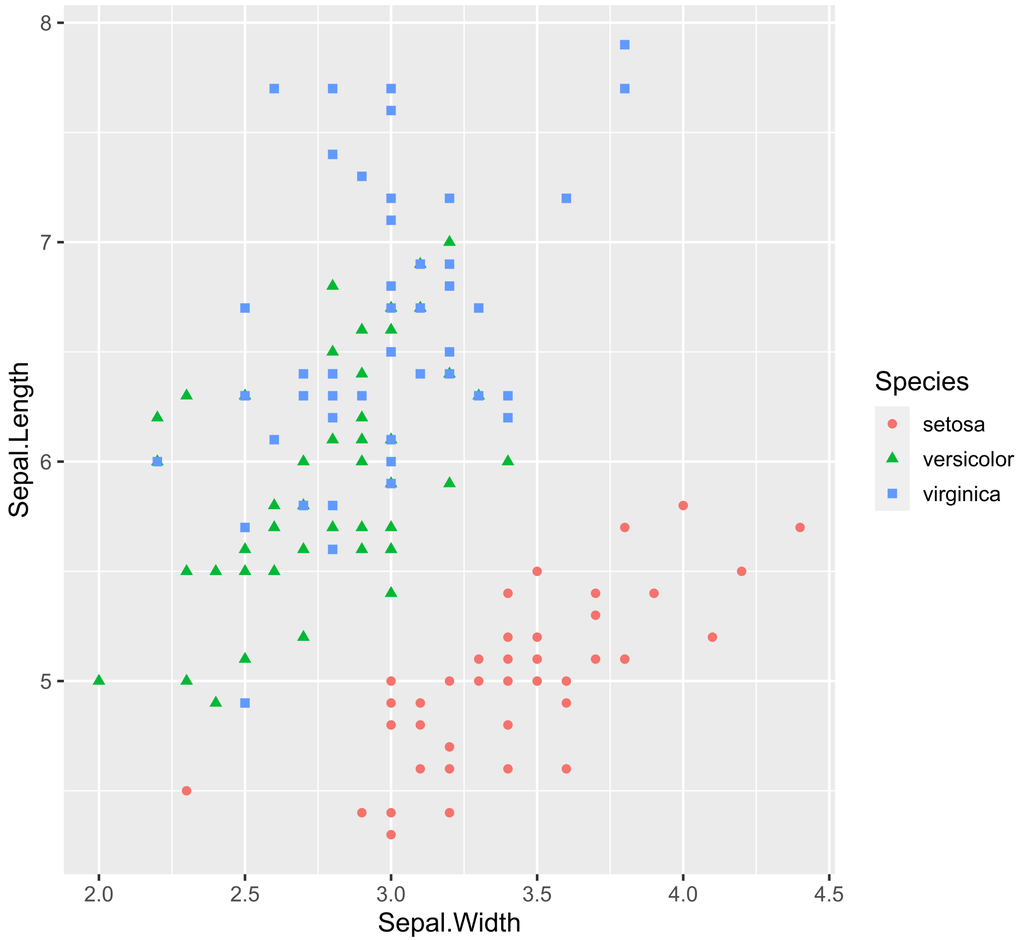


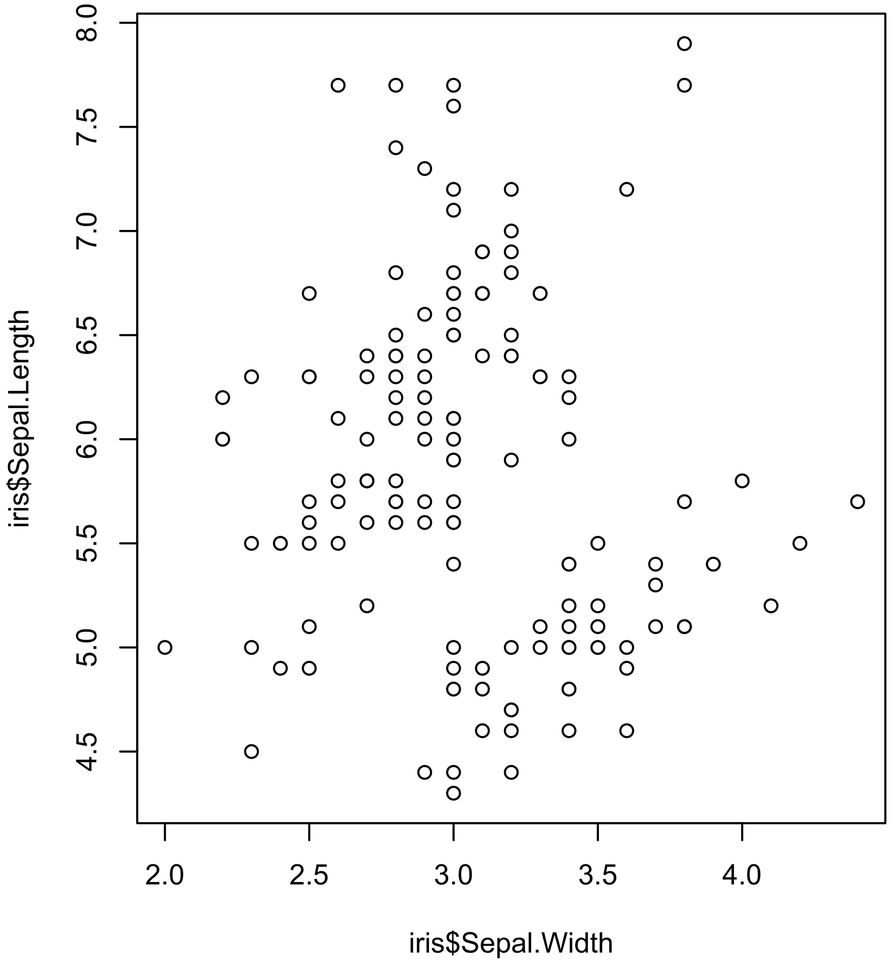
ggplot2


8
7
6
5
Sepal.Length
2.0
2.5
3.0
3.5
4.0
4.5
Sepal.Width

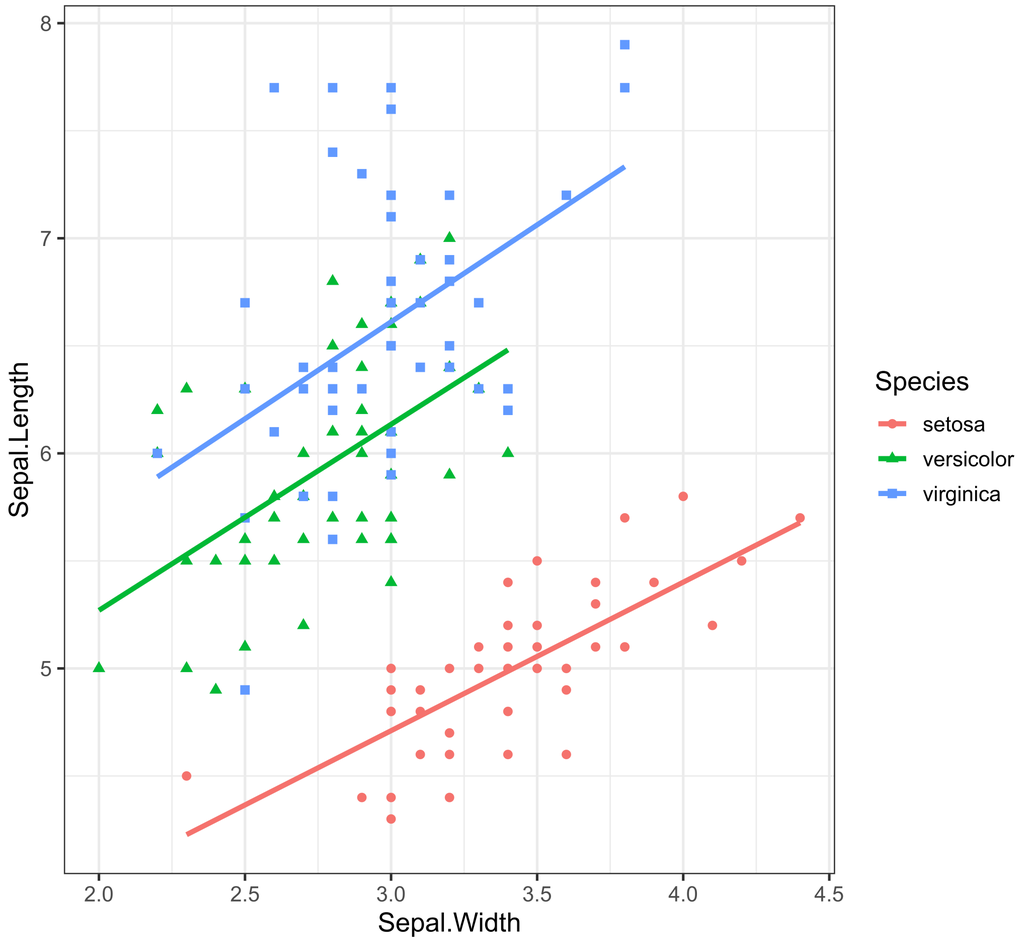

8.0
7.5
7.0
6.5
6.0
5.5
5.0
4.5
iris$Sepal.Length
2.0
2.5
3.0
3.5
4.0
iris$Sepal.Width

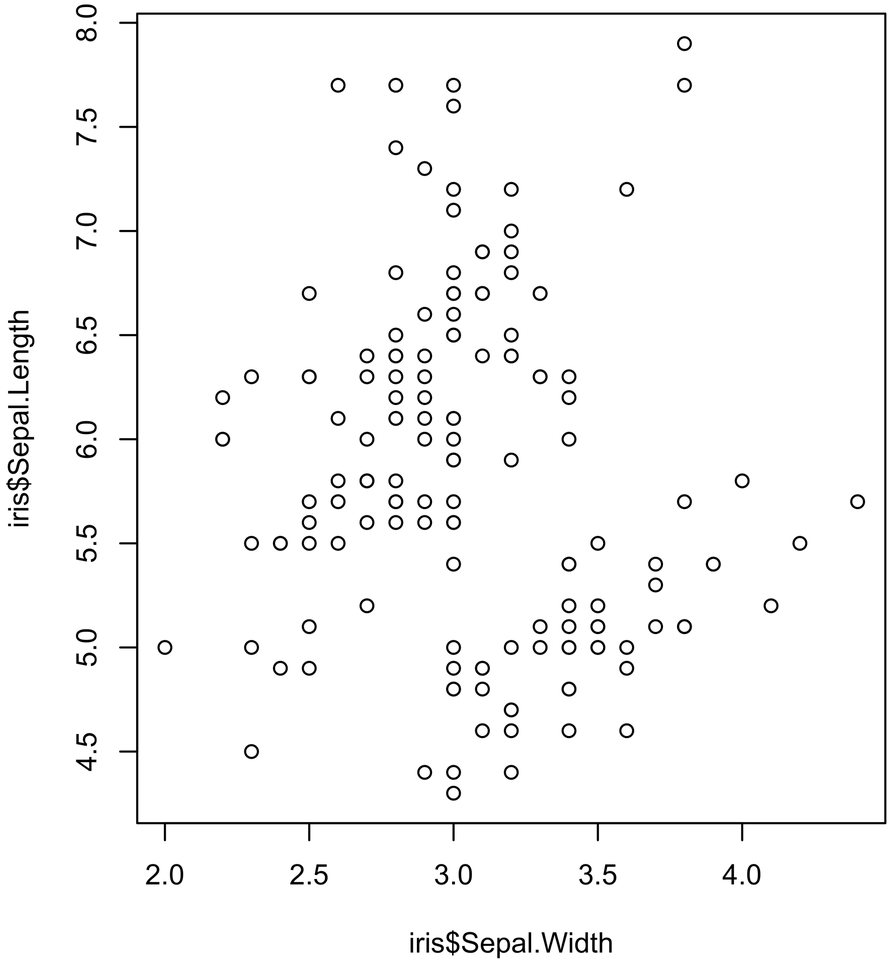
ggplot2


Sepal.Length
Sepal.Width
8
7
6
5
2.0
2.5
3.0
3.5
4.0
4.5
Species
setosa
versicolor
virginica

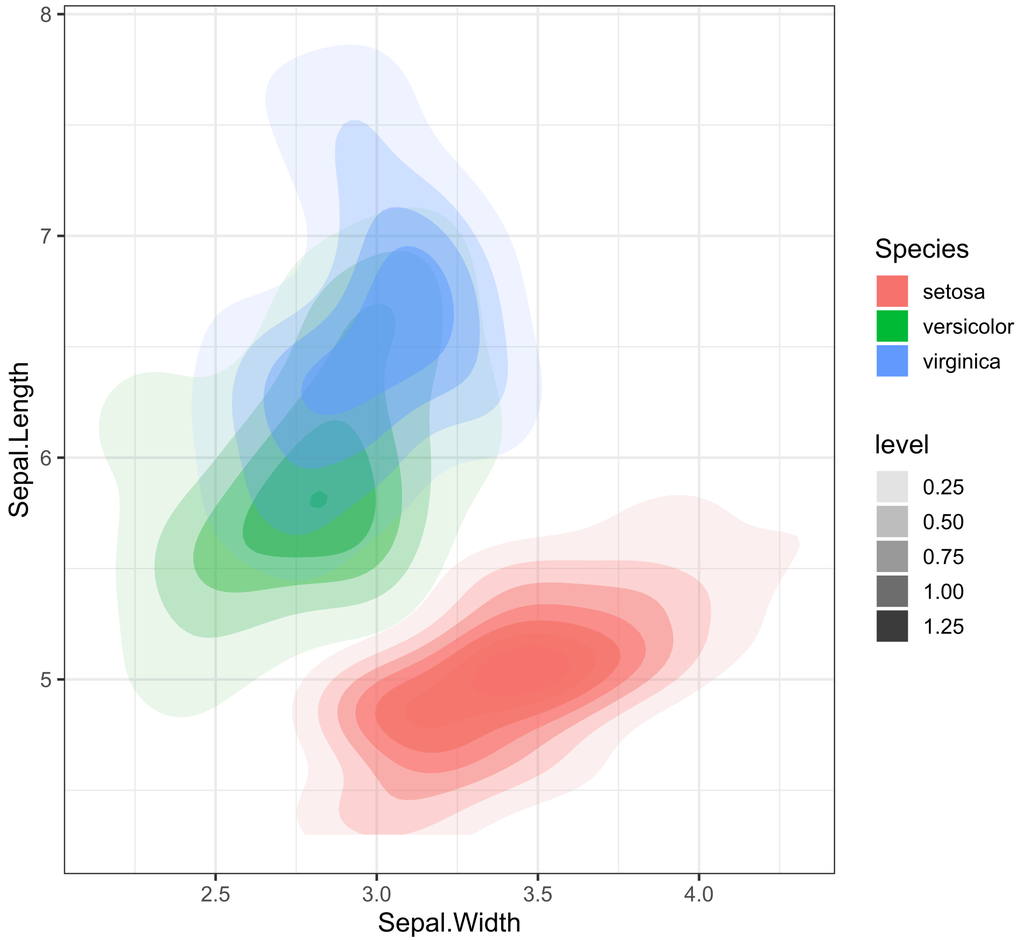

iris$Sepal.Length
iris$Sepal.Width
8.0
7.5
7.0
6.5
6.0
5.5
5.0
4.5
2.0
2.5
3.0
3.5
4.0

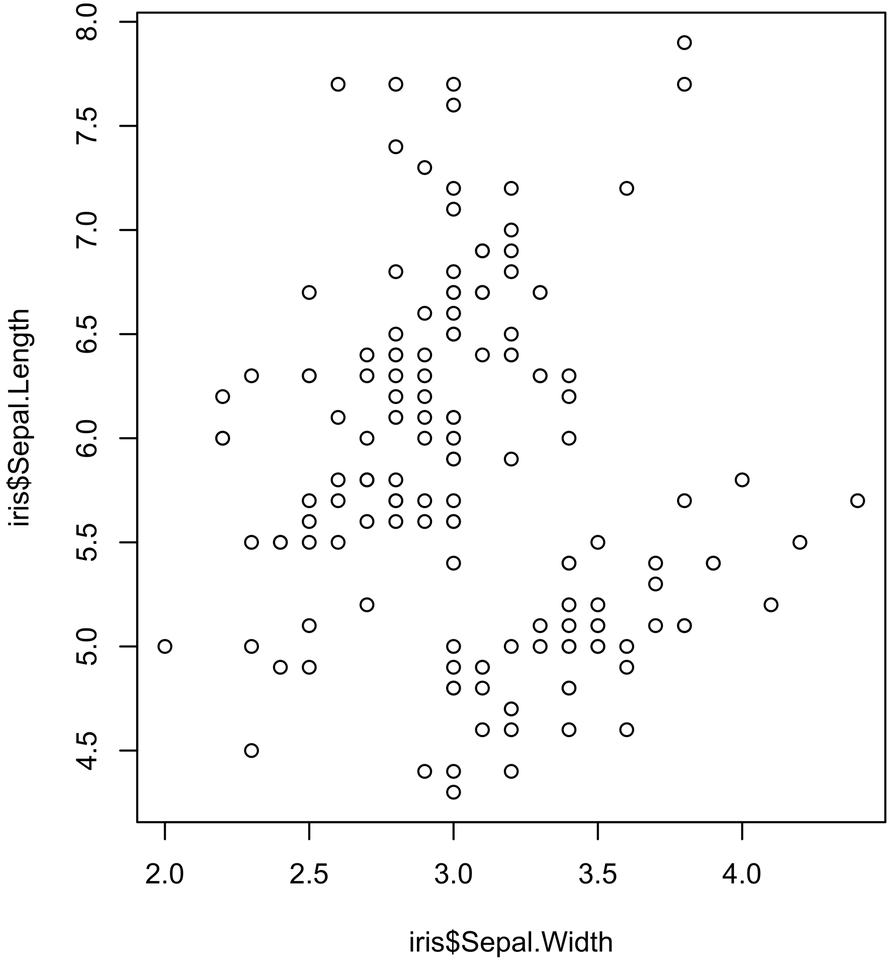
ggplot2


Sepal.Length
Sepal.Width
8
7
6
5
2.0
2.5
3.0
3.5
4.0
4.5
Species
setosa
versicolor
virginica

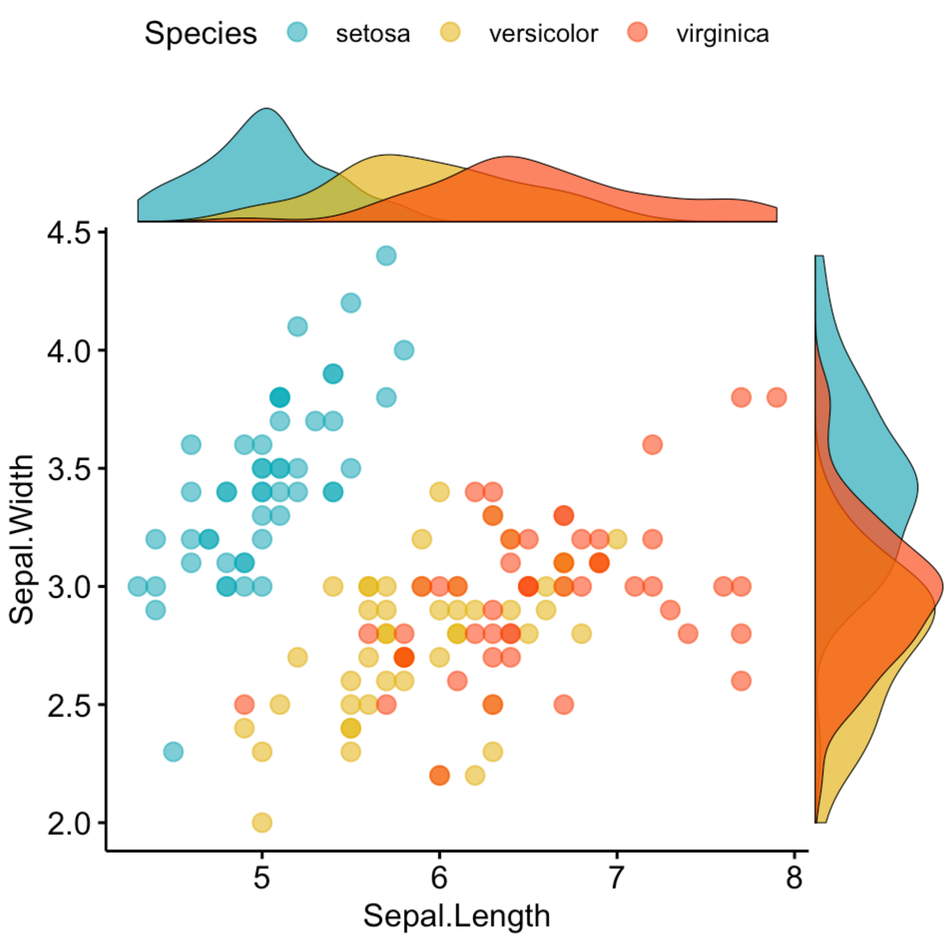

iris$Sepal.Length
iris$Sepal.Width
8.0
7.5
7.0
6.5
6.0
5.5
5.0
4.5
2.0
2.5
3.0
3.5
4.0

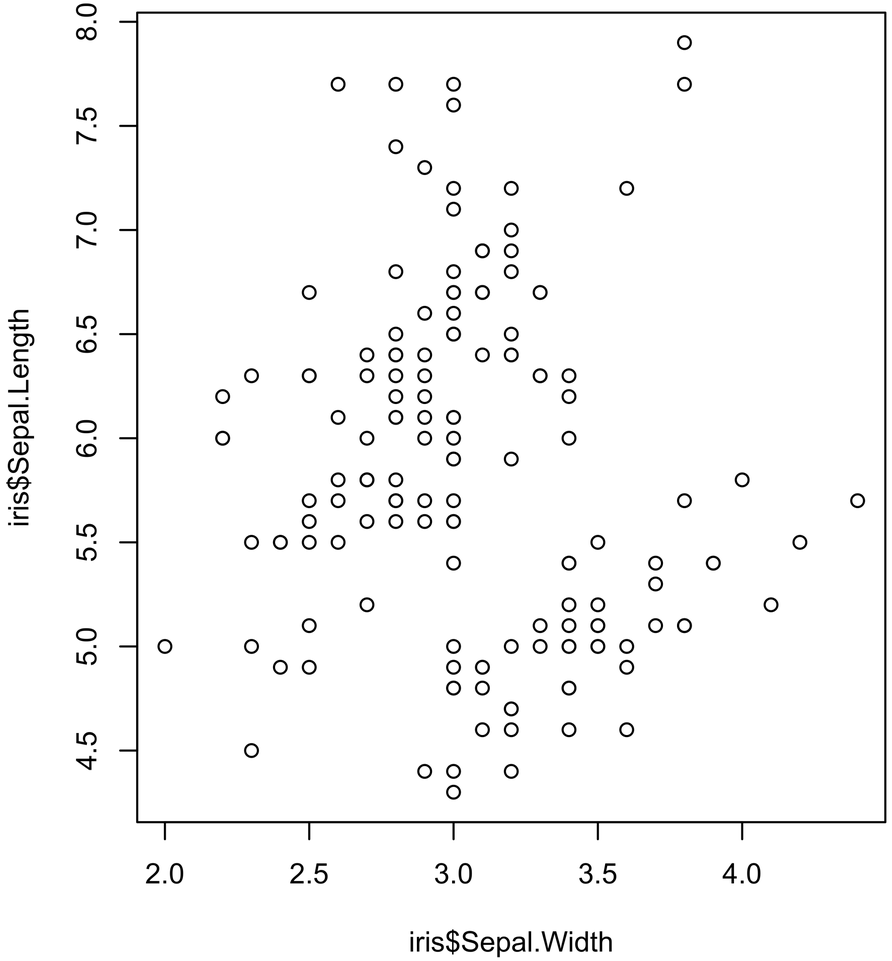
ggplot2


8
7
6
5
Sepal.Length
2.5
3.0
3.5
4.0
Sepal.Width
Species
setosa
versicolor
virginica
level
0.25
0.50
0.75
1.00
1.25


8.0
7.5
7.0
6.5
6.0
5.5
5.0
4.5
iris$Sepal.Length
2.0
2.5
3.0
3.5
4.0
iris$Sepal.Width

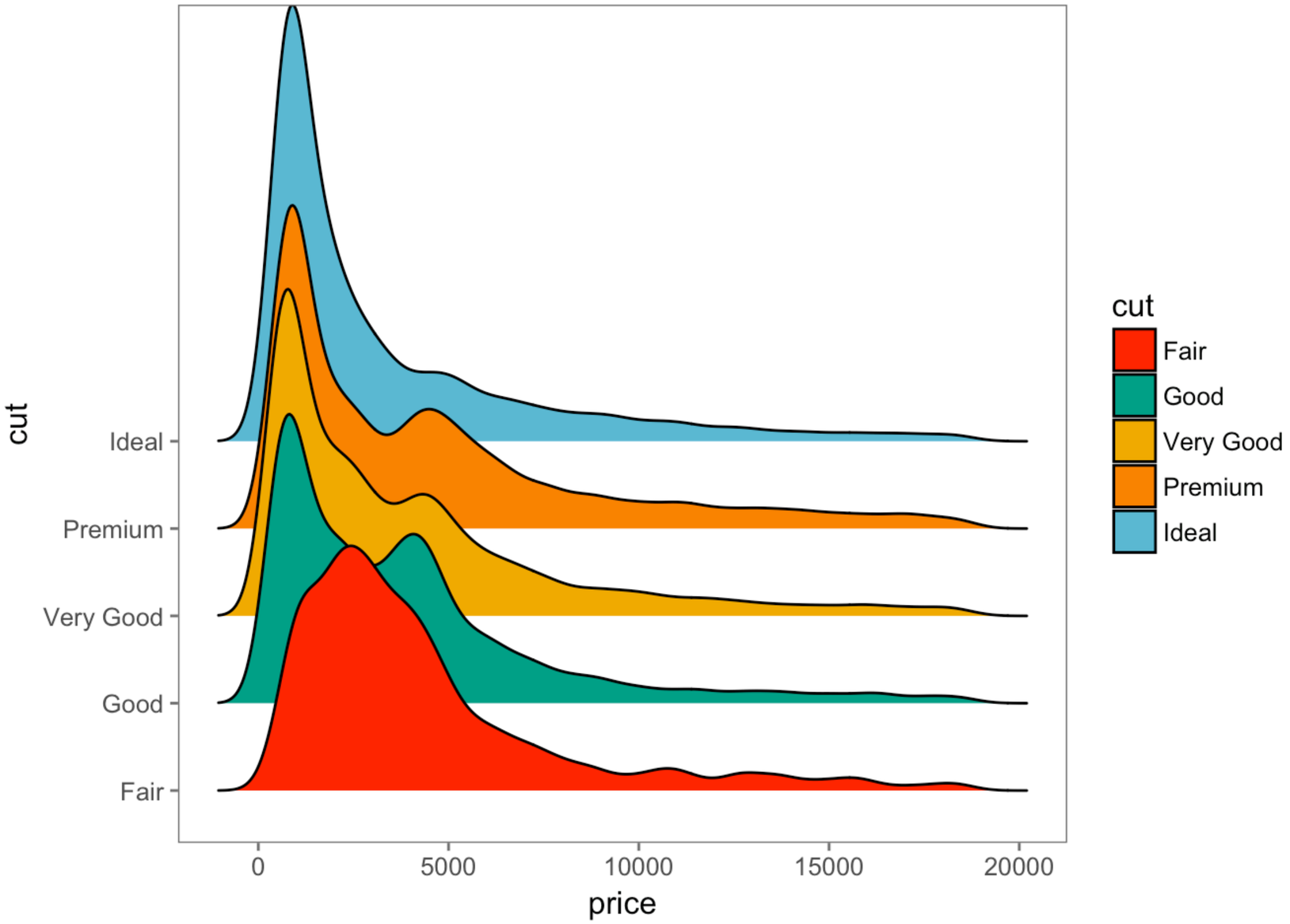
ggplot2

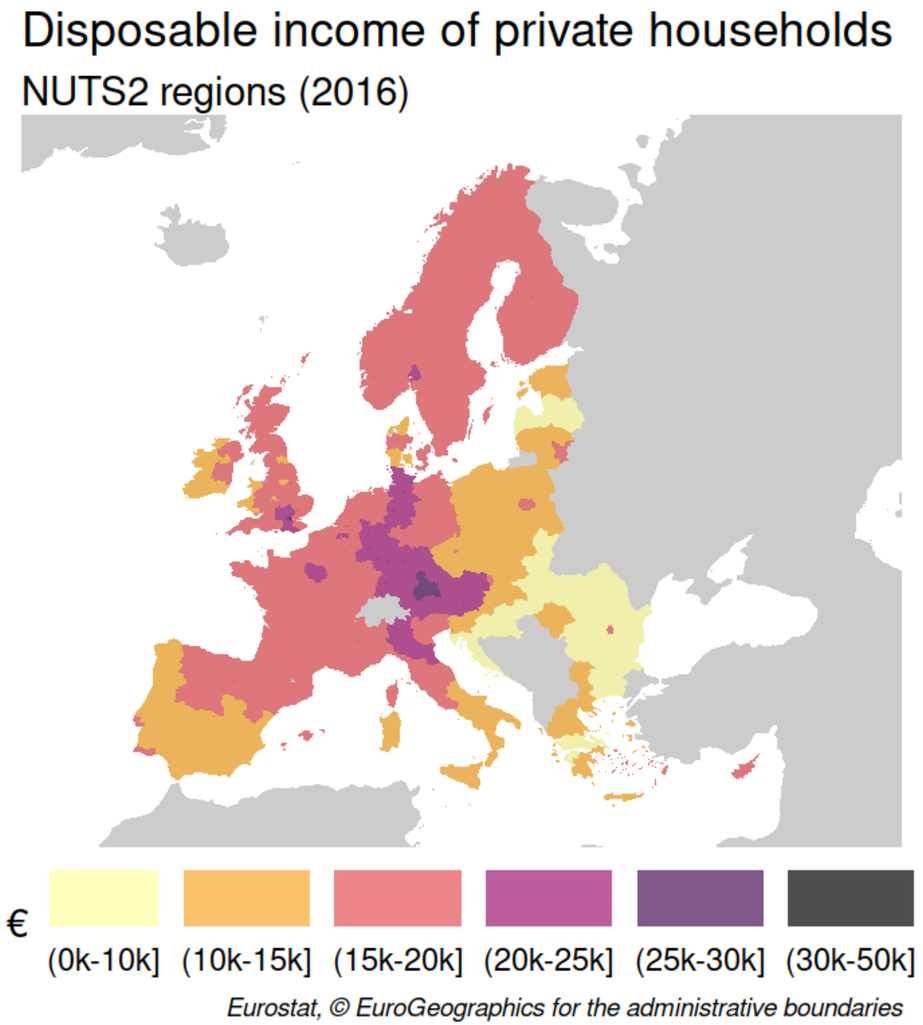

Species
setosa
versicolor
virginica
Sepal.Width
Sepal.Length
2.0
2.5
3.0
3.5
4.0
4.5
5
6
7
8

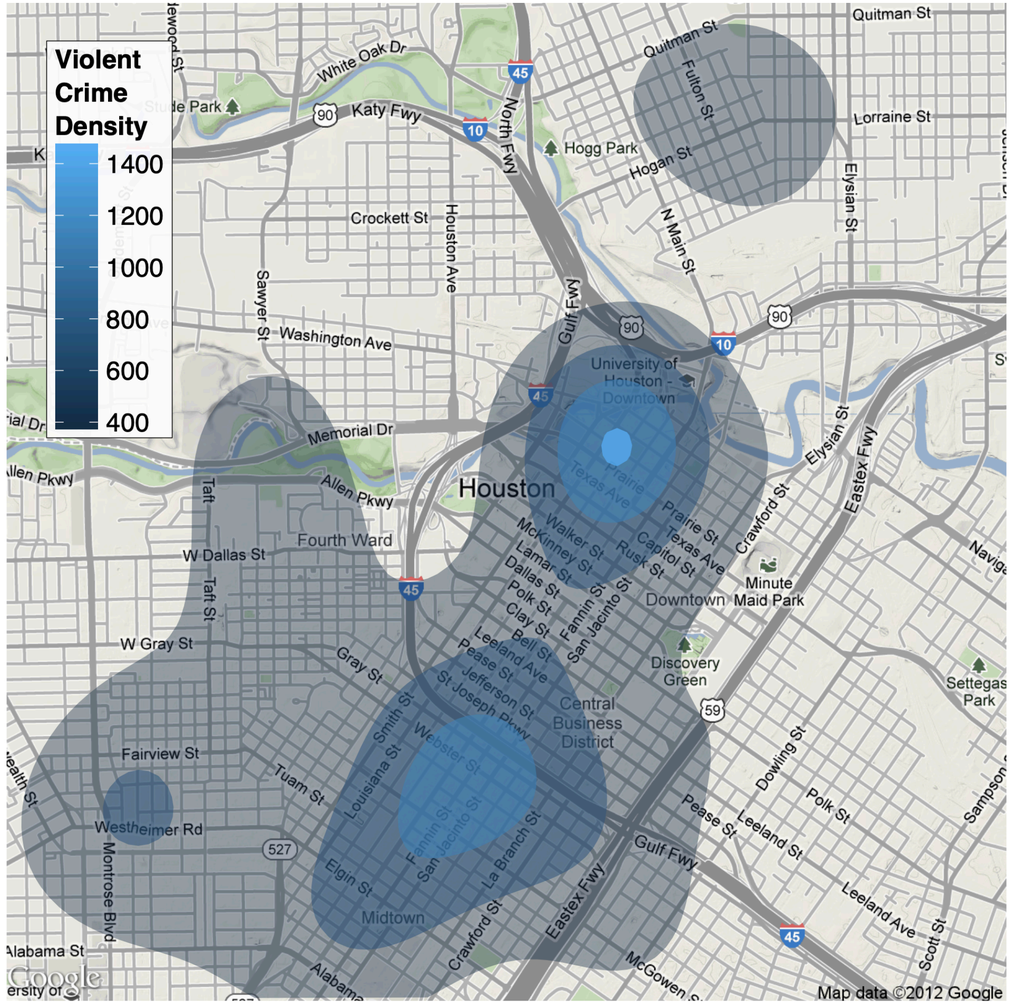

iris$Sepal.Length
iris$Sepal.Width
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0
2.0
2.5
3.0
3.5
4.0

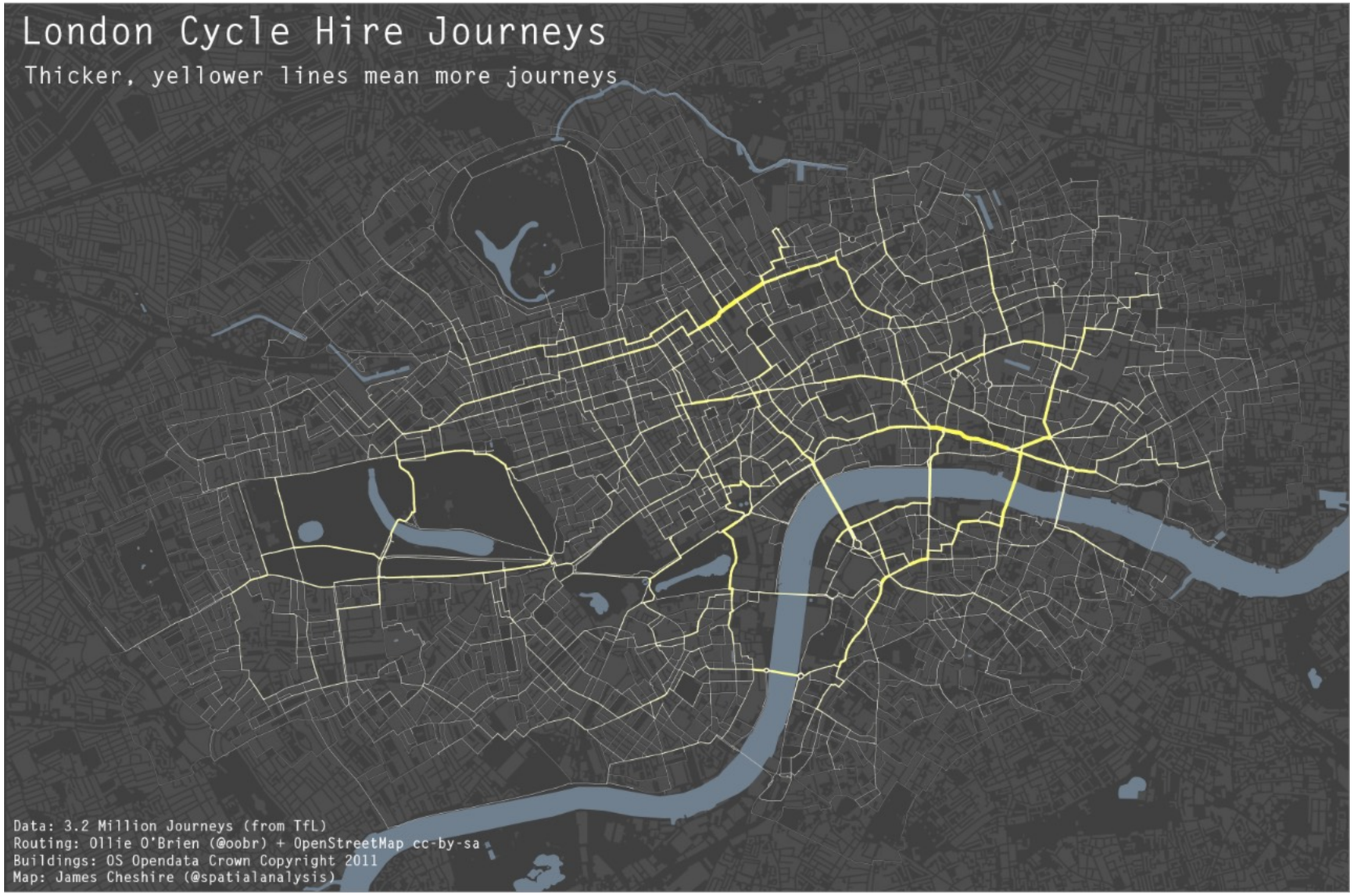
O maior valor de uma imagem está no momento em que nos obriga a perceber o que nunca esperaríamos ver.

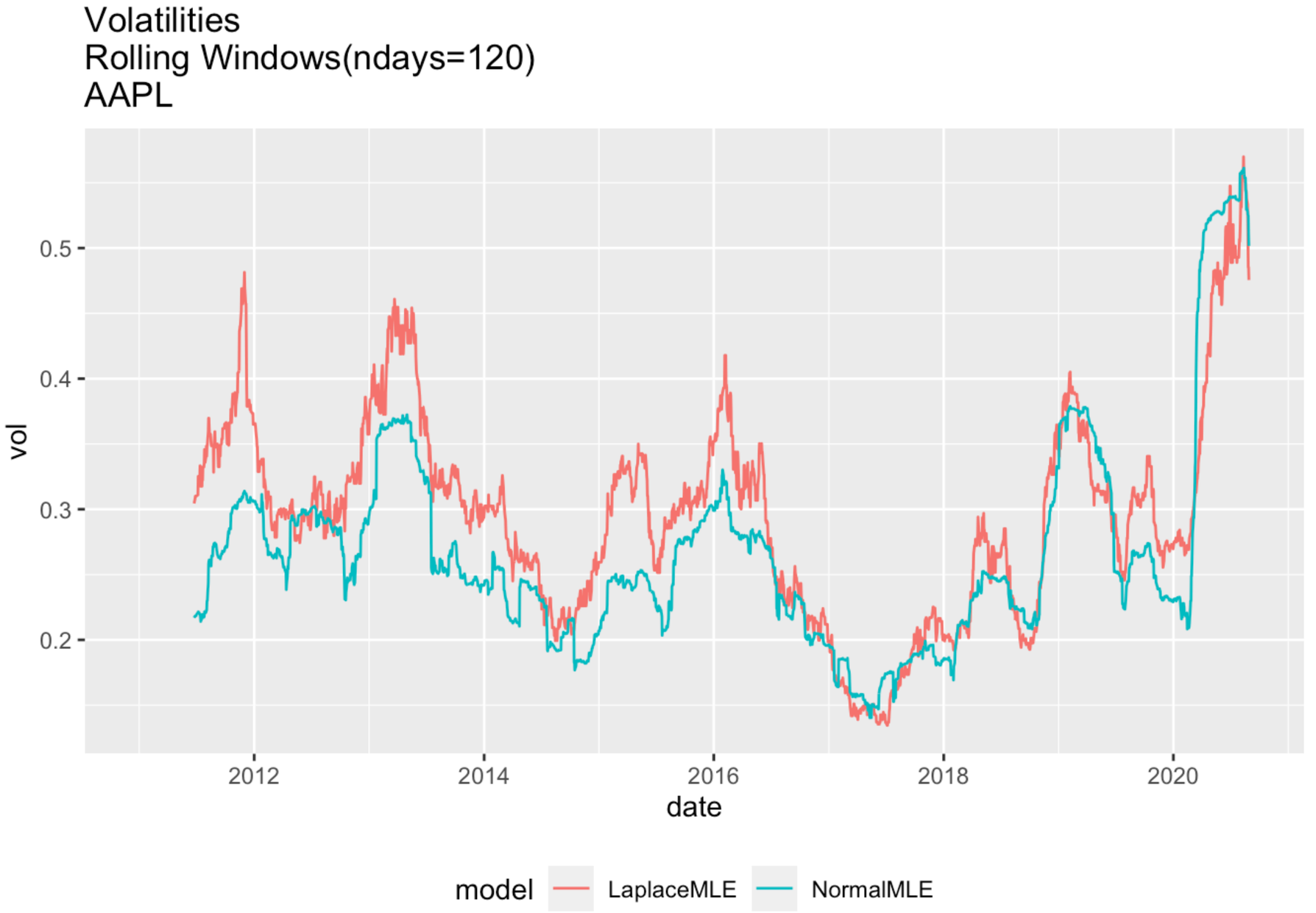
- John Tukey


cut
Ideal
Premium
Very Good
Good
Fair
0
5000
10000
15000
20000
price
cut
Fair
Good
Very Good
Premium
Ideal


Disposable income of private households
NUTS2 regions (2016)
€
(0k-10k]
(10k-15k]
(15k-20k]
(20k-25k]
(25k-30k]
(30k-50k]
Eurostat, © EuroGeographics for the administrative boundaries


Violent Crime Density
1400
1200
1000
800
600
400
Houston
Fourth Ward
Midtown
Downtown
Central Business District
University of Houston - Downtown
Minute Maid Park
Discovery Green
Hogg Park
Settegast Park
Memorial Dr
Allen Pkwy
W Dallas St
W Gray St
Fairview St
Westheimer Rd
Alabama St
Crockett St
Washington Ave
Katy Fwy
North Fwy
Gulf Fwy
Eastex Fwy
Quitman St
Lorraine St
Hogan St
Elysian St
N Main St
Houston Ave
Sawyer St
Taft St
Crawford St
Dowling St
Polk St
Leeland St
Leeland Ave
Pease St
Jefferson St
Gray St
Tuam St
Elgin St
Louisiana St
Smith St
Fannin St
San Jacinto St
La Branch St
Montrose Blvd
Scott St
Sampson St
Prairie St
Texas Ave
Capitol St
Rusk St
Walker St
McKinney St
Lamar St
Dallas St
Polk St
Clay St
Bell St
Leeland Ave
Pease St
Jefferson St
St Joseph Pkwy
White Oak Dr
Fulton St
Quitman St
Map data ©2012 Google


London Cycle Hire Journeys
Thicker, yellower lines mean more journeys
Data: 3.2 Million Journeys (from TfL)
Routing: Ollie O'Brien (@oobr) + OpenStreetMap cc-by-sa
Buildings: OS Opendata Crown Copyright 2011
Map: James Cheshire (@spatialanalysis)


Volatilities
Rolling Windows(ndays=120)
AAPL
0.5
0.4
0.3
0.2
vol
2012
2014
2016
2018
2020
date
model
LaplaceMLE
NormalMLE

Last Updated at (M/D/YYYY)
9/22/2022, 9:23 AM

Total Cases	Total Deaths	Total Vaccine Doses Administered
613,600,659	6,531,965	12,246,701,217

28-Day Cases	28-Day Deaths	28-Day Vaccine Doses Administered
14,939,622	51,171	145,559,885

Cases | Deaths by Country/Region/Sovereignty

Japan
28-Day: 3,052,150 | 6,147
Totals: 20,833,147 | 44,095

US
28-Day: 1,943,419 | 12,401
Totals: 95,876,628 | 1,055,216

Korea, South
28-Day: 1,834,019 | 1,677
Totals: 24,535,948 | 28,009

Russia
28-Day: 1,349,052 | 2,541
Totals: 20,343,828 | 378,628

Taiwan*
28-Day: 975,321 | 952
Totals: 6,132,794 | 10,702

Germany
28-Day: 932,708 | 2,478
Totals: 32,854,288 | 149,275

France
28-Day: 603,835 | 1,062
Totals: 35,228,668 | 155,864

Italy
28-Day: 497,216 | 1,672
Totals: 22,218,848 | 176,715

China
28-Day: 283,733 | 283
Totals: 2,715,281 | 15,166

Brazil
28-Day: 262,427 | 2,352
Totals: 34,592,027 | 685,428

Australia
28-Day: 203,762 | 1,286
Totals: 10,164,301 | 14,820

Turkey
28-Day: 180,534 | 668
Totals: 16,852,382 | 101,068

India
28-Day: 174,122 | 941
Totals: 44,553,043 | 528,429

Greece
28-Day: 163,190 | 675
Totals: 4,875,215 | 33,010

Chile
28-Day: 128,552 | 712
Totals: 4,590,421 | 60,966

United Kingdom
28-Day: 127,072 | 319
Totals: 23,803,458 | 207,189

Austria
28-Day: 118,861 | 97
Totals: 5,042,307 | 20,690

Poland
28-Day: 103,709 | 455
Totals: 6,263,616 | 117,412

Indonesia
28-Day: 86,185 | 510
Totals: 6,417,490 | 157,966

Serbia
28-Day: 83,610 | 339
Totals: 2,341,269 | 16,913

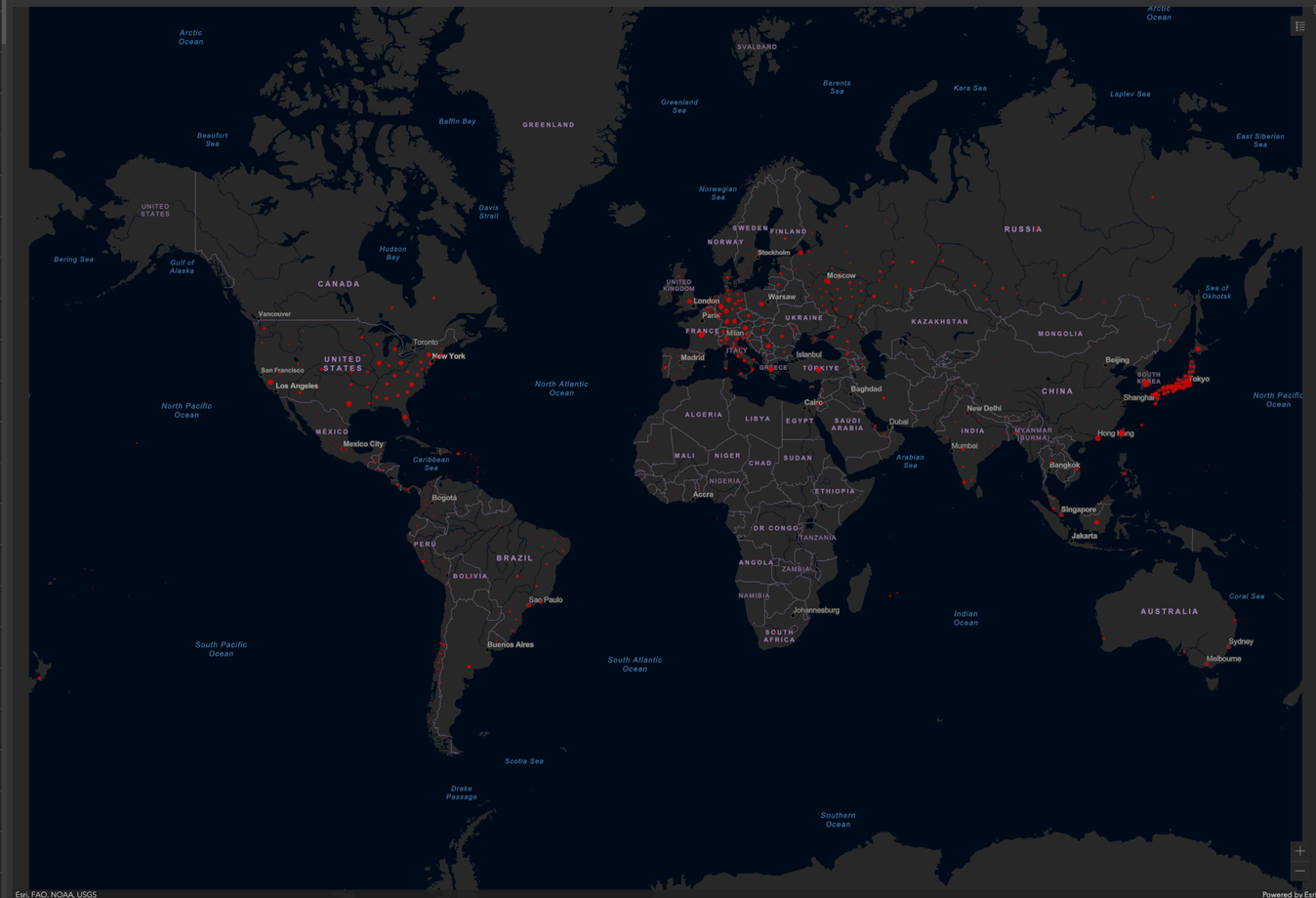
Mexico
28-Day: 81,696 | 724
Totals: 7,076,005 | 329,949

Ukraine
28-Day: 78,017 | 217
Totals: 5,394,128 | 116,774

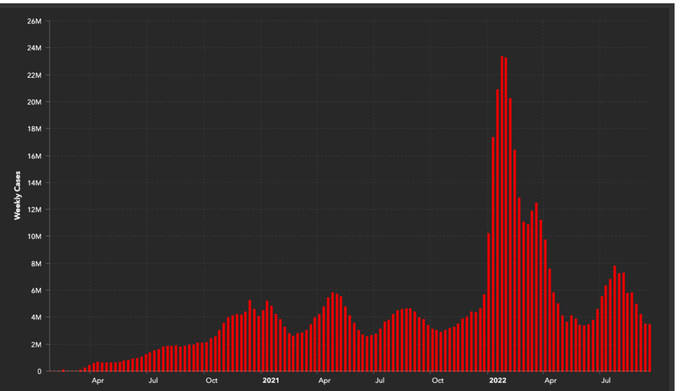
Canada
28-Day: 76,169 | 1,208
Totals: 4,251,898 | 45,067

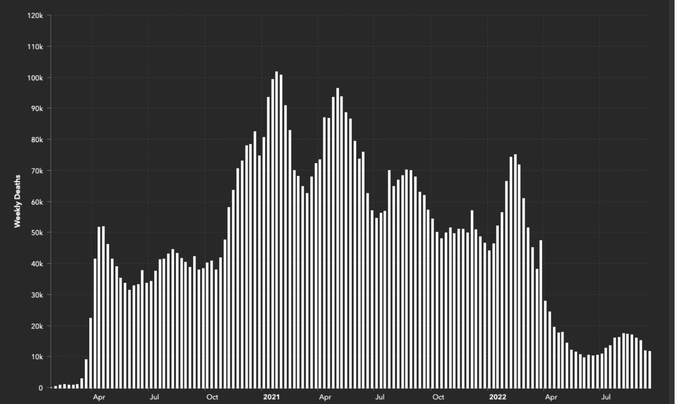
Vietnam
28-Day: 72,832 | 38

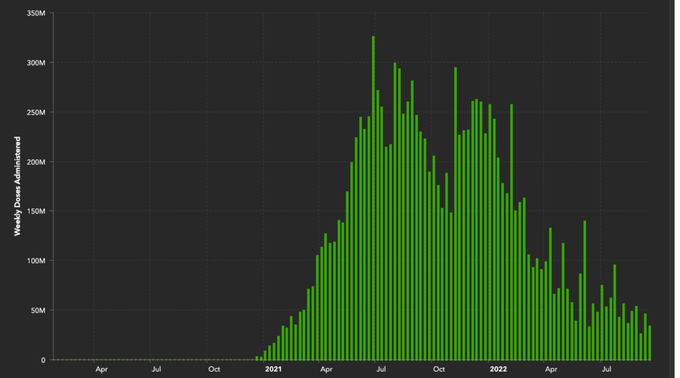
Admin0 | Admin1 | Admin2



28-Day | Totals | Incidence | Case-Fatality Ratio | Global Vaccinations | US Vaccinations | Terms of Use







Weekly | 28-Day

Fundamentos

Como iniciar um gráfico

ggplot2

- Para iniciarmos um gráfico via ggplot2 você deve indicar o seu conjuntos de dados

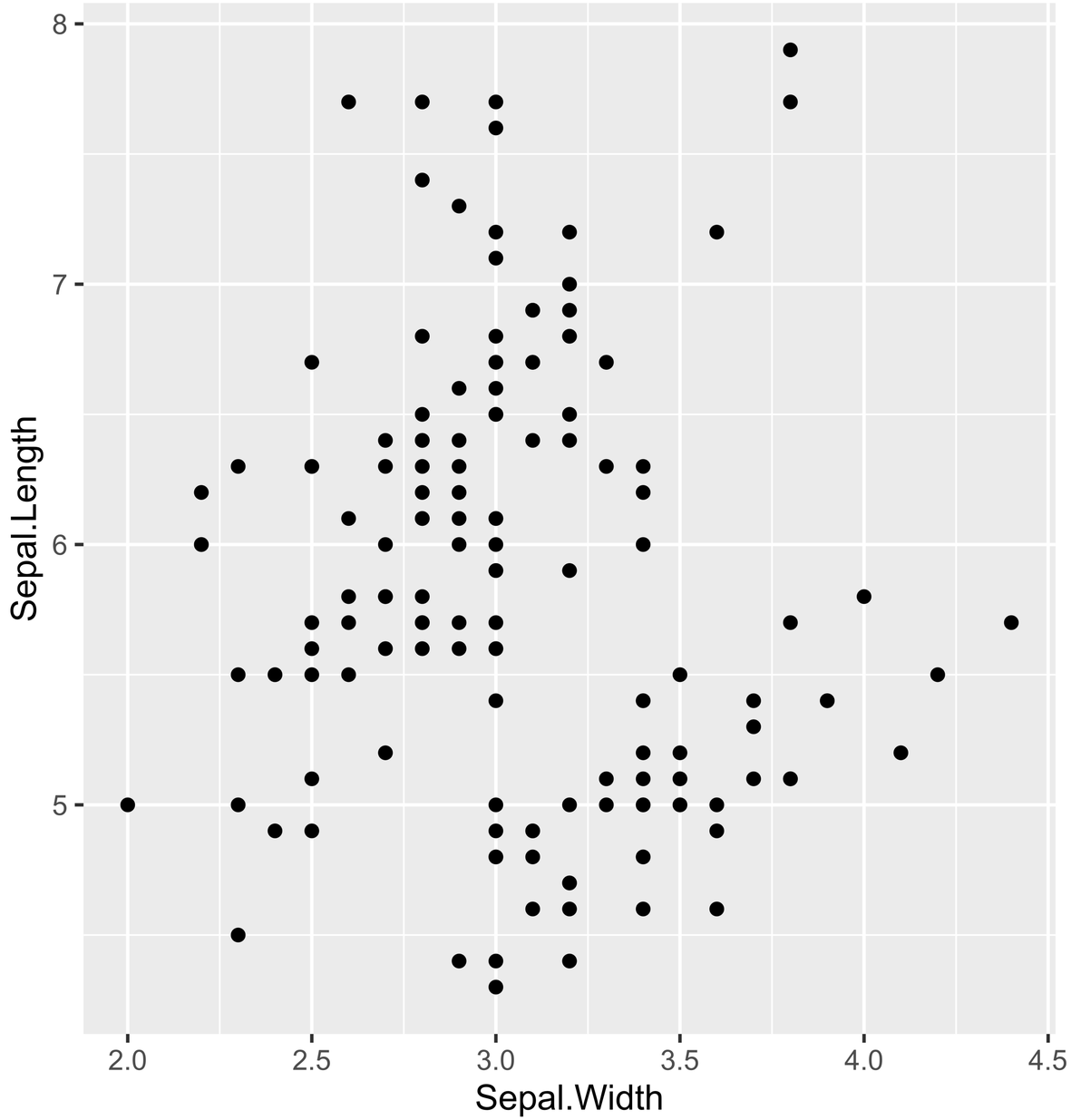
```
ggplot(data = iris)
```

- Note que o gráfico está vazio. Isso ocorre porque não lhe indicamos os elementos gráficos

ggplot2

- Nós adicionamos elementos por meio do "+"

```
ggplot(data = iris) +
 geom_point( mapping = aes(x = Sepal.Width, y = Sepal.Length) )
```


8
7
6
5
Sepal.Length
2.0
2.5
3.0
3.5
4.0
4.5
Sepal.Width

Dispersão

2 dimensões: contínuo x contínuo

ggplot2

- Nós adicionamos elementos por meio do "+"

```
ggplot(data = iris) +
  geom_point( mapping = aes(x = Sepal.Width, y = Sepal.Length) )
```

Variável x

Variável y

Geometria

Mapeamento

Estética

Dispersão

ggplot2

- Nós adicionamos elementos por meio do "+"

```
ggplot(iris) +
 geom_point( aes(Sepal.Width, Sepal.Length) )
```

mpg

```
> str(mpg)
tibble [234 × 11] (S3: tbl_df/tbl/data.frame)
 $ manufacturer: chr [1:234] "audi" "audi" "audi" "audi" ...
 $ model       : chr [1:234] "a4" "a4" "a4" "a4" ...
 $ displ       : num [1:234] 1.8 1.8 2 2 2.8 2.8 3.1 1.8 1.8 2 ...
 $ year        : int [1:234] 1999 1999 2008 2008 1999 1999 2008 1999 1999 2008 ...
 $ cyl         : int [1:234] 4 4 4 4 6 6 6 4 4 4 ...
 $ trans       : chr [1:234] "auto(l5)" "manual(m5)" "manual(m6)" "auto(av)" ...
 $ drv         : chr [1:234] "f" "f" "f" "f" ...
 $ cty         : int [1:234] 18 21 20 21 16 18 18 18 16 20 ...
 $ hwy         : int [1:234] 29 29 31 30 26 26 27 26 25 28 ...
 $ fl          : chr [1:234] "p" "p" "p" "p" ...
 $ class       : chr [1:234] "compact" "compact" "compact" "compact" ...
```

mpg

```
> str(mpg)
tibble [234 × 11] (S3: tbl_df/tbl/data.frame)
 $ manufacturer: chr [1:234] "audi" "audi" "audi" "audi" ...
 $ model       : chr [1:234] "a4" "a4" "a4" "a4" ...
 $ displ       : num [1:234] 1.8 1.8 2 2 2.8 2.8 3.1 1.8 1.8 2 ...
 $ year        : int [1:234] 1999 1999 2008 2008 1999 1999 2008 1999 1999 2008 ...
 $ cyl         : int [1:234] 4 4 4 4 6 6 6 4 4 4 ...
 $ trans       : chr [1:234] "auto(l5)" "manual(m5)" "manual(m6)" "auto(av)" ...
 $ drv         : chr [1:234] "f" "f" "f" "f" ...
 $ cty         : int [1:234] 18 21 20 21 16 18 18 18 16 20 ...
 $ hwy         : int [1:234] 29 29 31 30 26 26 27 26 25 28 ...
 $ fl          : chr [1:234] "p" "p" "p" "p" ...
 $ class       : chr [1:234] "compact" "compact" "compact" "compact" ...
```

Fabricante

mpg

```
> str(mpg)
tibble [234 × 11] (S3: tbl_df/tbl/data.frame)
 $ manufacturer: chr [1:234] "audi" "audi" "audi" "audi" ...
 $ model       : chr [1:234] "a4" "a4" "a4" "a4" ...
 $ displ       : num [1:234] 1.8 1.8 2 2 2.8 2.8 3.1 1.8 1.8 2 ...
 $ year        : int [1:234] 1999 1999 2008 2008 1999 1999 2008 1999 1999 2008 ...
 $ cyl         : int [1:234] 4 4 4 4 6 6 6 4 4 4 ...
 $ trans       : chr [1:234] "auto(l5)" "manual(m5)" "manual(m6)" "auto(av)" ...
 $ drv         : chr [1:234] "f" "f" "f" "f" ...
 $ cty         : int [1:234] 18 21 20 21 16 18 18 18 16 20 ...
 $ hwy         : int [1:234] 29 29 31 30 26 26 27 26 25 28 ...
 $ fl          : chr [1:234] "p" "p" "p" "p" ...
 $ class       : chr [1:234] "compact" "compact" "compact" "compact" ...
```

Modelo

mpg

```
> str(mpg)
tibble [234 × 11] (S3: tbl_df/tbl/data.frame)
 $ manufacturer: chr [1:234] "audi" "audi" "audi" "audi" ...
 $ model       : chr [1:234] "a4" "a4" "a4" "a4" ...
 $ displ       : num [1:234] 1.8 1.8 2 2 2.8 2.8 3.1 1.8 1.8 2 ...
 $ year        : int [1:234] 1999 1999 2008 2008 1999 1999 2008 1999 1999 2008 ...
 $ cyl         : int [1:234] 4 4 4 4 6 6 6 4 4 4 ...
 $ trans       : chr [1:234] "auto(l5)" "manual(m5)" "manual(m6)" "auto(av)" ...
 $ drv         : chr [1:234] "f" "f" "f" "f" ...
 $ cty         : int [1:234] 18 21 20 21 16 18 18 18 16 20 ...
 $ hwy         : int [1:234] 29 29 31 30 26 26 27 26 25 28 ...
 $ fl          : chr [1:234] "p" "p" "p" "p" ...
 $ class       : chr [1:234] "compact" "compact" "compact" "compact" ...
```

Cilindradas

mpg

```
> str(mpg)
tibble [234 × 11] (S3: tbl_df/tbl/data.frame)
 $ manufacturer: chr [1:234] "audi" "audi" "audi" "audi" ...
 $ model       : chr [1:234] "a4" "a4" "a4" "a4" ...
 $ displ       : num [1:234] 1.8 1.8 2 2 2.8 2.8 3.1 1.8 1.8 2 ...
 $ year        : int [1:234] 1999 1999 2008 2008 1999 1999 2008 1999 1999 2008 ...
 $ cyl         : int [1:234] 4 4 4 4 6 6 6 4 4 4 ...
 $ trans       : chr [1:234] "auto(l5)" "manual(m5)" "manual(m6)" "auto(av)" ...
 $ drv         : chr [1:234] "f" "f" "f" "f" ...
 $ cty         : int [1:234] 18 21 20 21 16 18 18 18 16 20 ...
 $ hwy         : int [1:234] 29 29 31 30 26 26 27 26 25 28 ...
 $ fl          : chr [1:234] "p" "p" "p" "p" ...
 $ class       : chr [1:234] "compact" "compact" "compact" "compact" ...
```

Ano

mpg

```
> str(mpg)
tibble [234 × 11] (S3: tbl_df/tbl/data.frame)
 $ manufacturer: chr [1:234] "audi" "audi" "audi" "audi" ...
 $ model       : chr [1:234] "a4" "a4" "a4" "a4" ...
 $ displ       : num [1:234] 1.8 1.8 2 2 2.8 2.8 3.1 1.8 1.8 2 ...
 $ year        : int [1:234] 1999 1999 2008 2008 1999 1999 2008 1999 1999 2008 ...
 $ cyl         : int [1:234] 4 4 4 4 6 6 6 4 4 4 ...
 $ trans       : chr [1:234] "auto(l5)" "manual(m5)" "manual(m6)" "auto(av)" ...
 $ drv         : chr [1:234] "f" "f" "f" "f" ...
 $ cty         : int [1:234] 18 21 20 21 16 18 18 18 16 20 ...
 $ hwy         : int [1:234] 29 29 31 30 26 26 27 26 25 28 ...
 $ fl          : chr [1:234] "p" "p" "p" "p" ...
 $ class       : chr [1:234] "compact" "compact" "compact" "compact" ...
```

Cilindros

mpg

```
> str(mpg)
tibble [234 × 11] (S3: tbl_df/tbl/data.frame)
 $ manufacturer: chr [1:234] "audi" "audi" "audi" "audi" ...
 $ model       : chr [1:234] "a4" "a4" "a4" "a4" ...
 $ displ       : num [1:234] 1.8 1.8 2 2 2.8 2.8 3.1 1.8 1.8 2 ...
 $ year        : int [1:234] 1999 1999 2008 2008 1999 1999 2008 1999 1999 2008 ...
 $ cyl         : int [1:234] 4 4 4 4 6 6 6 4 4 4 ...
 $ trans       : chr [1:234] "auto(l5)" "manual(m5)" "manual(m6)" "auto(av)" ...
 $ drv         : chr [1:234] "f" "f" "f" "f" ...
 $ cty         : int [1:234] 18 21 20 21 16 18 18 18 16 20 ...
 $ hwy         : int [1:234] 29 29 31 30 26 26 27 26 25 28 ...
 $ fl          : chr [1:234] "p" "p" "p" "p" ...
 $ class       : chr [1:234] "compact" "compact" "compact" "compact" ...
```

Transmissão

mpg

```
> str(mpg)
tibble [234 × 11] (S3: tbl_df/tbl/data.frame)
 $ manufacturer: chr [1:234] "audi" "audi" "audi" "audi" ...
 $ model       : chr [1:234] "a4" "a4" "a4" "a4" ...
 $ displ       : num [1:234] 1.8 1.8 2 2 2.8 2.8 3.1 1.8 1.8 2 ...
 $ year        : int [1:234] 1999 1999 2008 2008 1999 1999 2008 1999 1999 2008 ...
 $ cyl         : int [1:234] 4 4 4 4 6 6 6 4 4 4 ...
 $ trans       : chr [1:234] "auto(l5)" "manual(m5)" "manual(m6)" "auto(av)" ...
 $ drv         : chr [1:234] "f" "f" "f" "f" ...
 $ cty         : int [1:234] 18 21 20 21 16 18 18 18 16 20 ...
 $ hwy         : int [1:234] 29 29 31 30 26 26 27 26 25 28 ...
 $ fl          : chr [1:234] "p" "p" "p" "p" ...
 $ class       : chr [1:234] "compact" "compact" "compact" "compact" ...
```

Tração

mpg

```
> str(mpg)
tibble [234 × 11] (S3: tbl_df/tbl/data.frame)
 $ manufacturer: chr [1:234] "audi" "audi" "audi" "audi" ...
 $ model       : chr [1:234] "a4" "a4" "a4" "a4" ...
 $ displ       : num [1:234] 1.8 1.8 2 2 2.8 2.8 3.1 1.8 1.8 2 ...
 $ year        : int [1:234] 1999 1999 2008 2008 1999 1999 2008 1999 1999 2008 ...
 $ cyl         : int [1:234] 4 4 4 4 6 6 6 4 4 4 ...
 $ trans       : chr [1:234] "auto(l5)" "manual(m5)" "manual(m6)" "auto(av)" ...
 $ drv         : chr [1:234] "f" "f" "f" "f" ...
 $ cty         : int [1:234] 18 21 20 21 16 18 18 18 16 20 ...
 $ hwy         : int [1:234] 29 29 31 30 26 26 27 26 25 28 ...
 $ fl          : chr [1:234] "p" "p" "p" "p" ...
 $ class       : chr [1:234] "compact" "compact" "compact" "compact" ...
```

Autonomia na cidade

mpg

```
> str(mpg)
tibble [234 × 11] (S3: tbl_df/tbl/data.frame)
 $ manufacturer: chr [1:234] "audi" "audi" "audi" "audi" ...
 $ model       : chr [1:234] "a4" "a4" "a4" "a4" ...
 $ displ       : num [1:234] 1.8 1.8 2 2 2.8 2.8 3.1 1.8 1.8 2 ...
 $ year        : int [1:234] 1999 1999 2008 2008 1999 1999 2008 1999 1999 2008 ...
 $ cyl         : int [1:234] 4 4 4 4 6 6 6 4 4 4 ...
 $ trans       : chr [1:234] "auto(l5)" "manual(m5)" "manual(m6)" "auto(av)" ...
 $ drv         : chr [1:234] "f" "f" "f" "f" ...
 $ cty         : int [1:234] 18 21 20 21 16 18 18 18 16 20 ...
 $ hwy         : int [1:234] 29 29 31 30 26 26 27 26 25 28 ...
 $ fl          : chr [1:234] "p" "p" "p" "p" ...
 $ class       : chr [1:234] "compact" "compact" "compact" "compact" ...
```

Autonomia na estrada

mpg

```
> str(mpg)
tibble [234 × 11] (S3: tbl_df/tbl/data.frame)
 $ manufacturer: chr [1:234] "audi" "audi" "audi" "audi" ...
 $ model       : chr [1:234] "a4" "a4" "a4" "a4" ...
 $ displ       : num [1:234] 1.8 1.8 2 2 2.8 2.8 3.1 1.8 1.8 2 ...
 $ year        : int [1:234] 1999 1999 2008 2008 1999 1999 2008 1999 1999 2008 ...
 $ cyl         : int [1:234] 4 4 4 4 6 6 6 4 4 4 ...
 $ trans       : chr [1:234] "auto(l5)" "manual(m5)" "manual(m6)" "auto(av)" ...
 $ drv         : chr [1:234] "f" "f" "f" "f" ...
 $ cty         : int [1:234] 18 21 20 21 16 18 18 18 16 20 ...
 $ hwy         : int [1:234] 29 29 31 30 26 26 27 26 25 28 ...
 $ fl          : chr [1:234] "p" "p" "p" "p" ...
 $ class       : chr [1:234] "compact" "compact" "compact" "compact" ...
```

Tipo do combustível

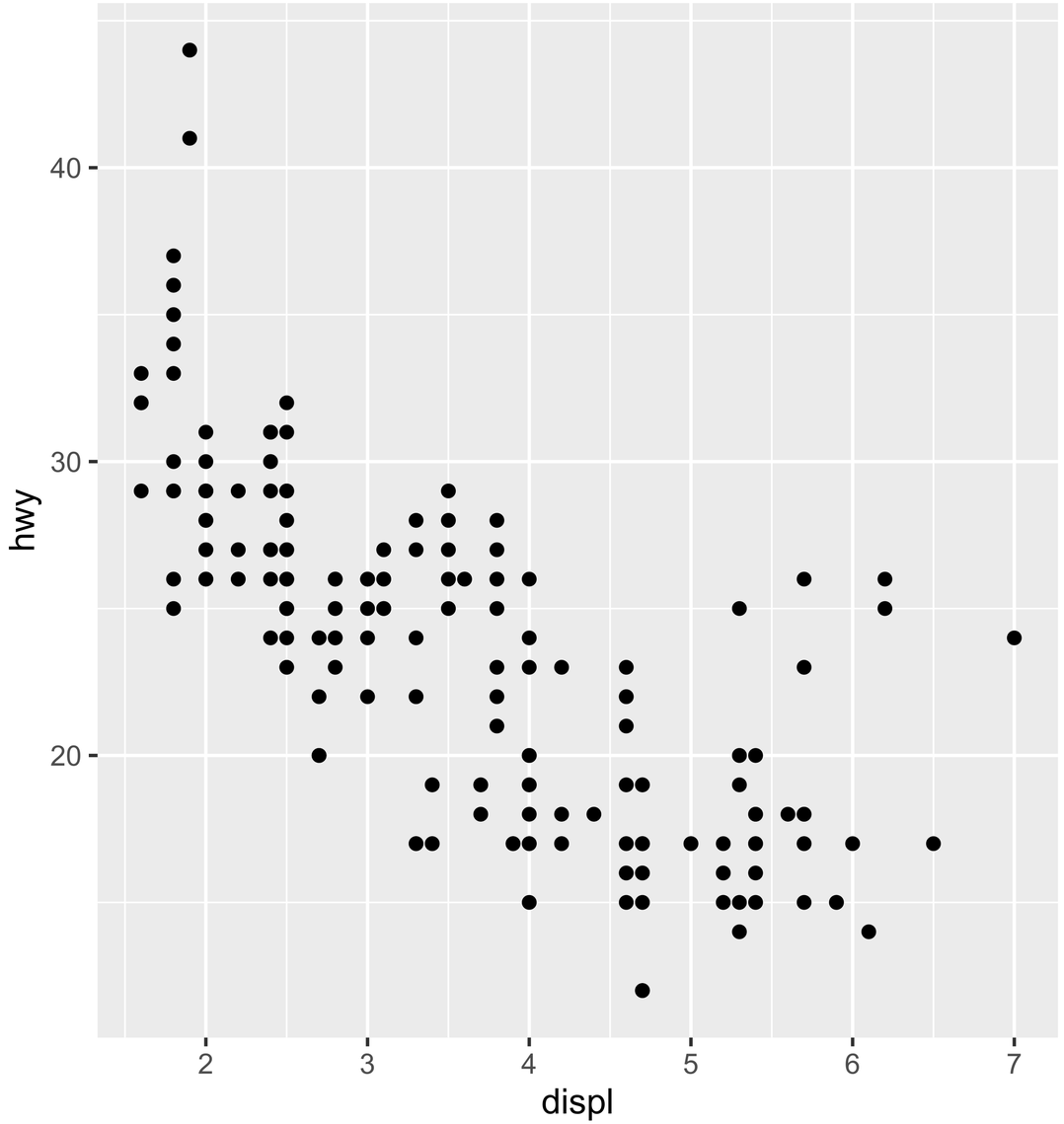
mpg

```
> str(mpg)
tibble [234 × 11] (S3: tbl_df/tbl/data.frame)
 $ manufacturer: chr [1:234] "audi" "audi" "audi" "audi" ...
 $ model       : chr [1:234] "a4" "a4" "a4" "a4" ...
 $ displ       : num [1:234] 1.8 1.8 2 2 2.8 2.8 3.1 1.8 1.8 2 ...
 $ year        : int [1:234] 1999 1999 2008 2008 1999 1999 2008 1999 1999 2008 ...
 $ cyl         : int [1:234] 4 4 4 4 6 6 6 4 4 4 ...
 $ trans       : chr [1:234] "auto(l5)" "manual(m5)" "manual(m6)" "auto(av)" ...
 $ drv         : chr [1:234] "f" "f" "f" "f" ...
 $ cty         : int [1:234] 18 21 20 21 16 18 18 18 16 20 ...
 $ hwy         : int [1:234] 29 29 31 30 26 26 27 26 25 28 ...
 $ fl          : chr [1:234] "p" "p" "p" "p" ...
 $ class       : chr [1:234] "compact" "compact" "compact" "compact" ...
```

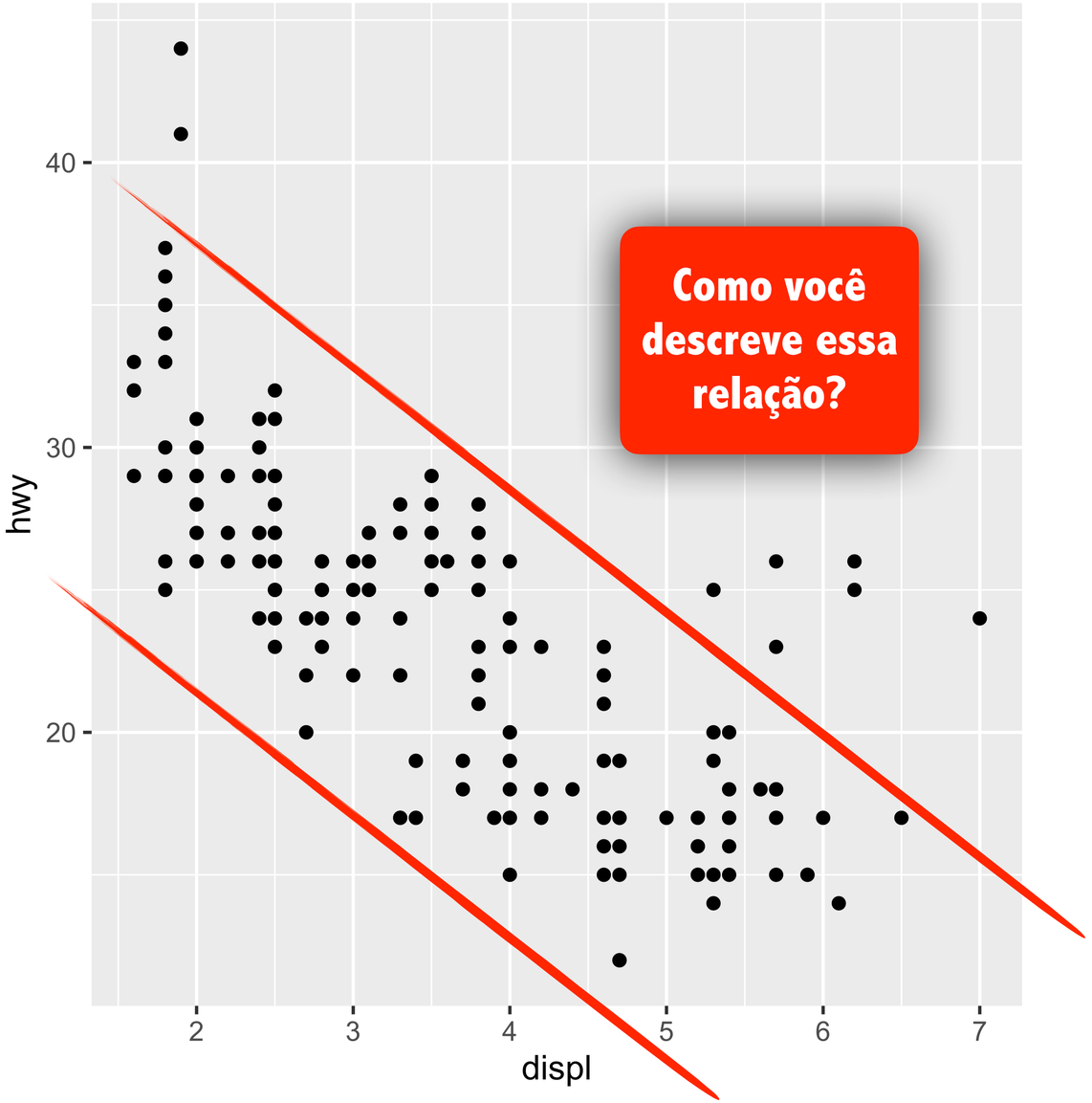
Classe

mpg

```
> head(mpg, 20)
# A tibble: 20 × 11
   manufacturer model              displ  year   cyl trans      drv     cty   hwy fl    class
   <chr>        <chr>              <dbl> <int> <int> <chr>      <chr> <int> <int> <chr> <chr>
 1 audi         a4                   1.8  1999     4 auto(l5)   f        18    29 p     compact
 2 audi         a4                   1.8  1999     4 manual(m5) f        21    29 p     compact
 3 audi         a4                   2    2008     4 manual(m6) f        20    31 p     compact
 4 audi         a4                   2    2008     4 auto(av)   f        21    30 p     compact
 5 audi         a4                   2.8  1999     6 auto(l5)   f        16    26 p     compact
 6 audi         a4                   2.8  1999     6 manual(m5) f        18    26 p     compact
 7 audi         a4                   3.1  2008     6 auto(av)   f        18    27 p     compact
 8 audi         a4 quattro           1.8  1999     4 manual(m5) 4        18    26 p     compact
 9 audi         a4 quattro           1.8  1999     4 auto(l5)   4        16    25 p     compact
10 audi         a4 quattro           2    2008     4 manual(m6) 4        20    28 p     compact
11 audi         a4 quattro           2    2008     4 auto(s6)   4        19    27 p     compact
12 audi         a4 quattro           2.8  1999     6 auto(l5)   4        15    25 p     compact
13 audi         a4 quattro           2.8  1999     6 manual(m5) 4        17    25 p     compact
14 audi         a4 quattro           3.1  2008     6 auto(s6)   4        17    25 p     compact
15 audi         a4 quattro           3.1  2008     6 manual(m6) 4        15    25 p     compact
16 audi         a6 quattro           2.8  1999     6 auto(l5)   4        15    24 p     midsize
17 audi         a6 quattro           3.1  2008     6 auto(s6)   4        17    25 p     midsize
18 audi         a6 quattro           4.2  2008     8 auto(s6)   4        16    23 p     midsize
19 chevrolet    c1500 suburban 2wd   5.3  2008     8 auto(l4)   r        14    20 r     suv
20 chevrolet    c1500 suburban 2wd   5.3  2008     8 auto(l4)   r        11    15 e     suv
>
```



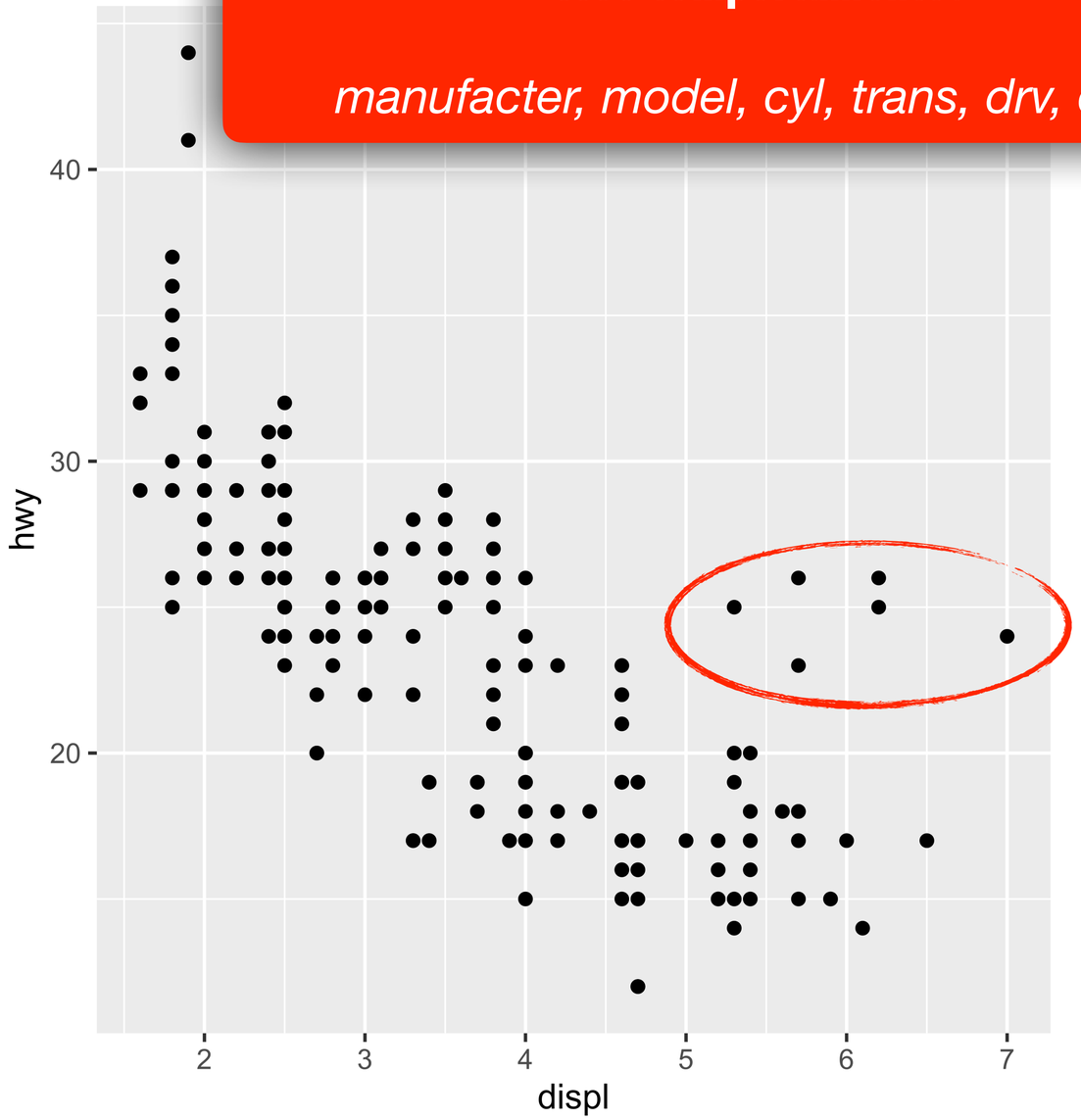

```
ggplot(mpg) +
 geom_point( aes(x = displ, y = hwy) )
```




```
ggplot(mpg) +
  geom_point( aes(x = displ, y = hwy) )
```

Que outras variáveis podem nos ajudar a compreender esse comportamento?

manufacter, model, cyl, trans, drv, class



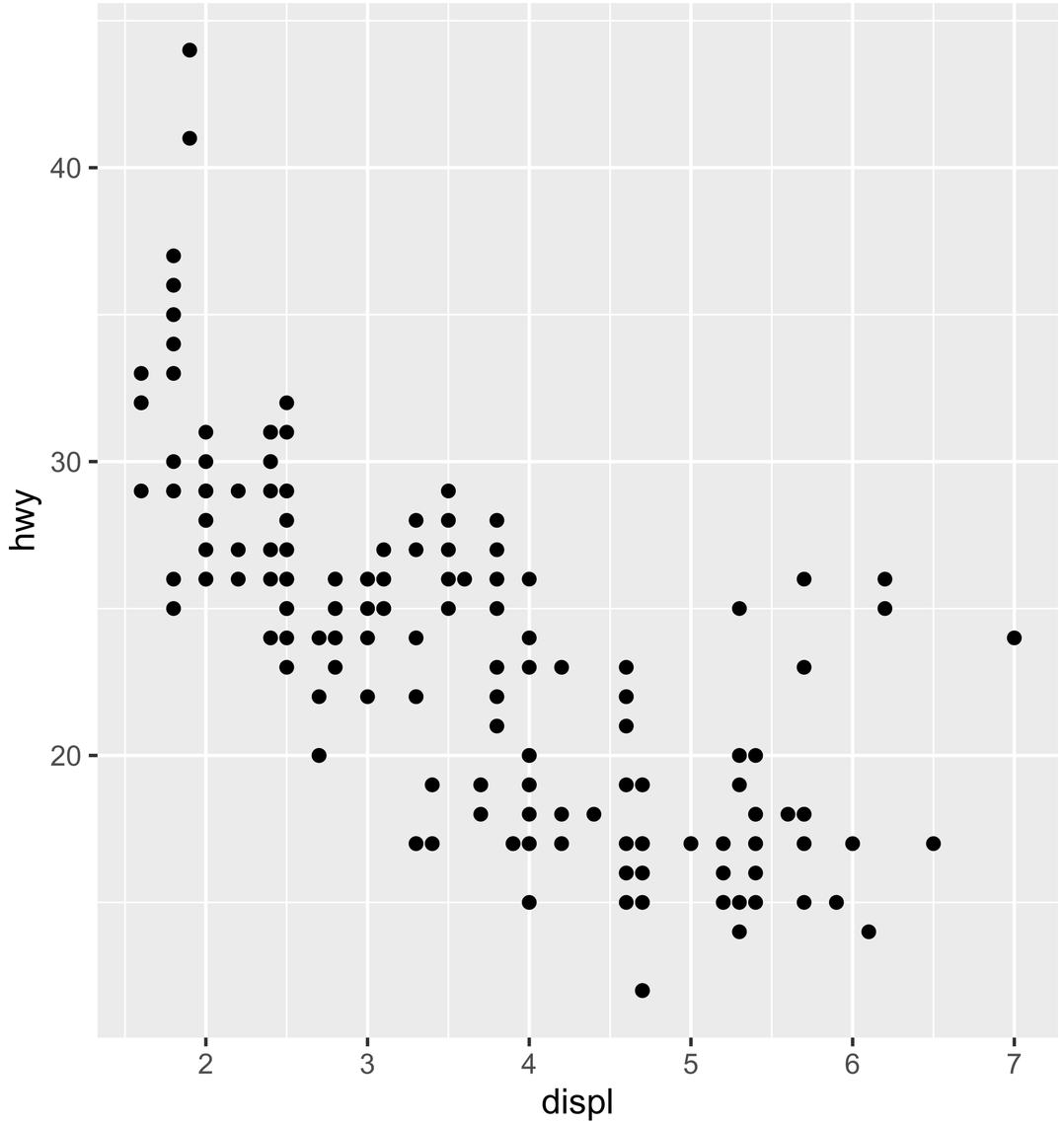

```
ggplot(mpg) +
  geom_point( aes(x = displ, y = hwy) )
```

Aesthetics

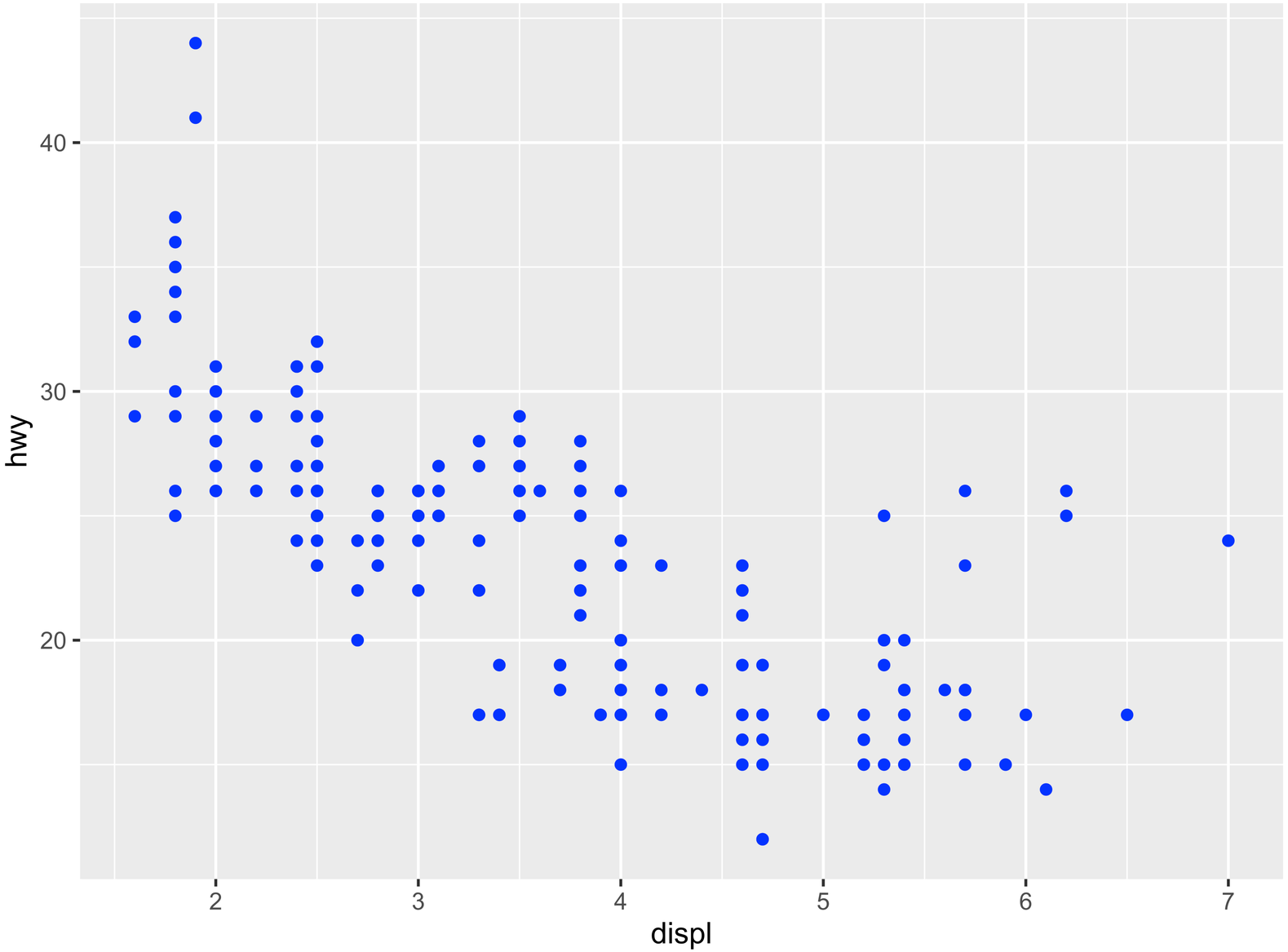
Mapeamento estético

Aesthetics

Características visuais que podem ser mapeadas aos dados



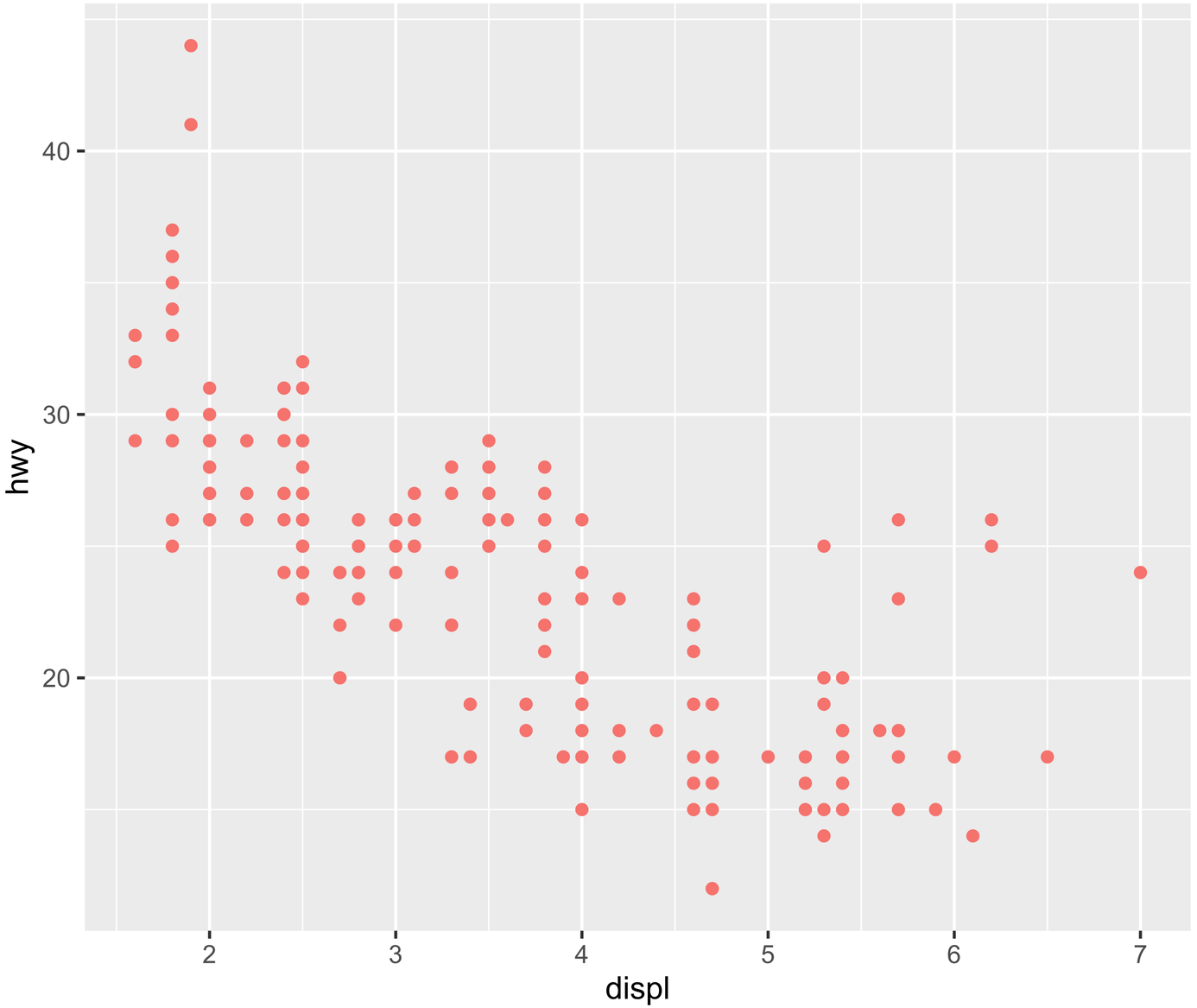

```
ggplot(mpg) +
 geom_point( aes(x = displ, y = hwy) )
```



Parâmetro estético

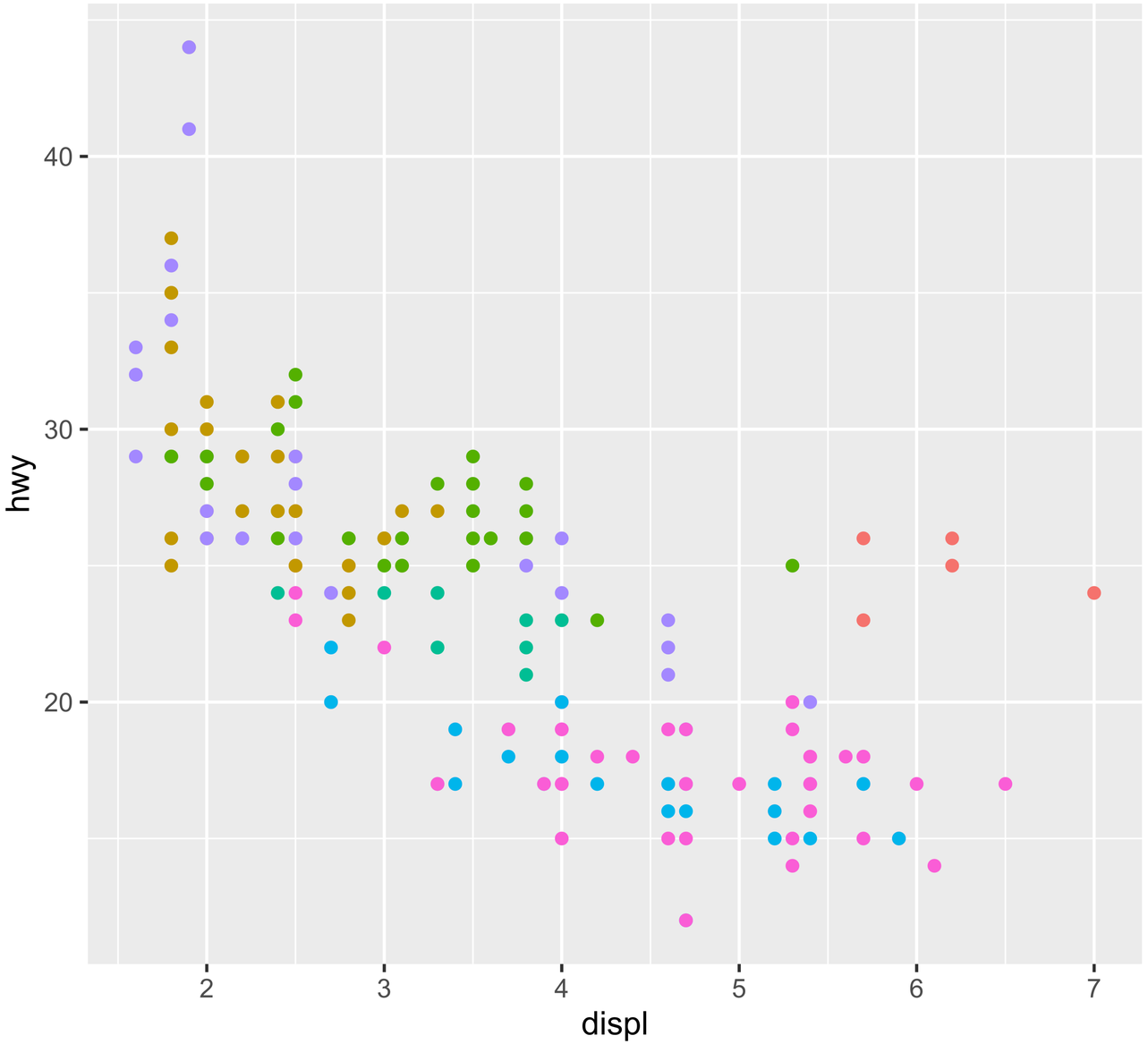
Valor estético

```
ggplot(mpg) +  geom_point( aes(x = displ, y = hwy), color = "blue" )
```



Atributo da geometria e não de variáveis

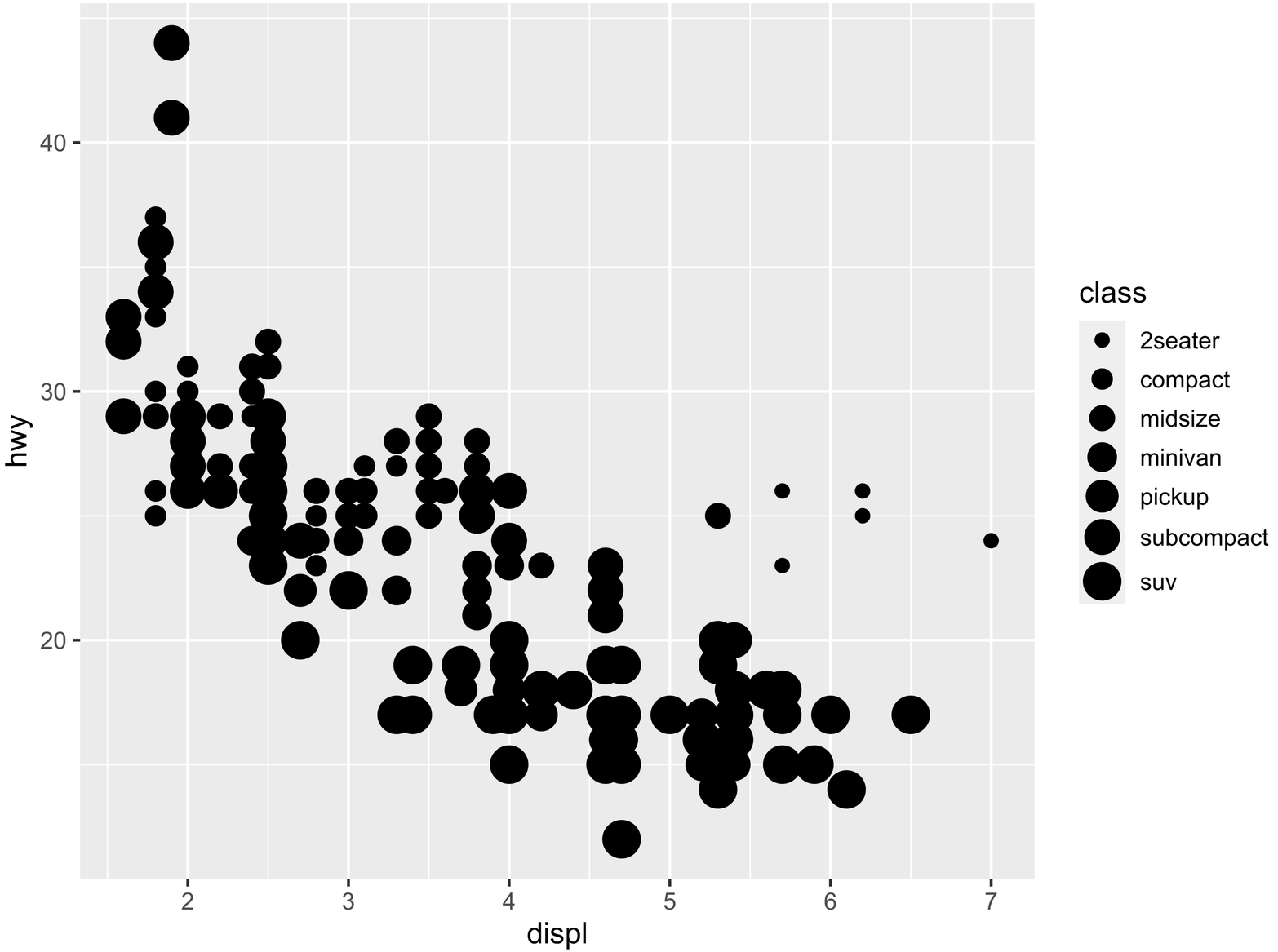
```
ggplot(mpg) +  geom_point( aes(x = displ, y = hwy, color = "blue") )
```



Legenda adicionada automaticamente

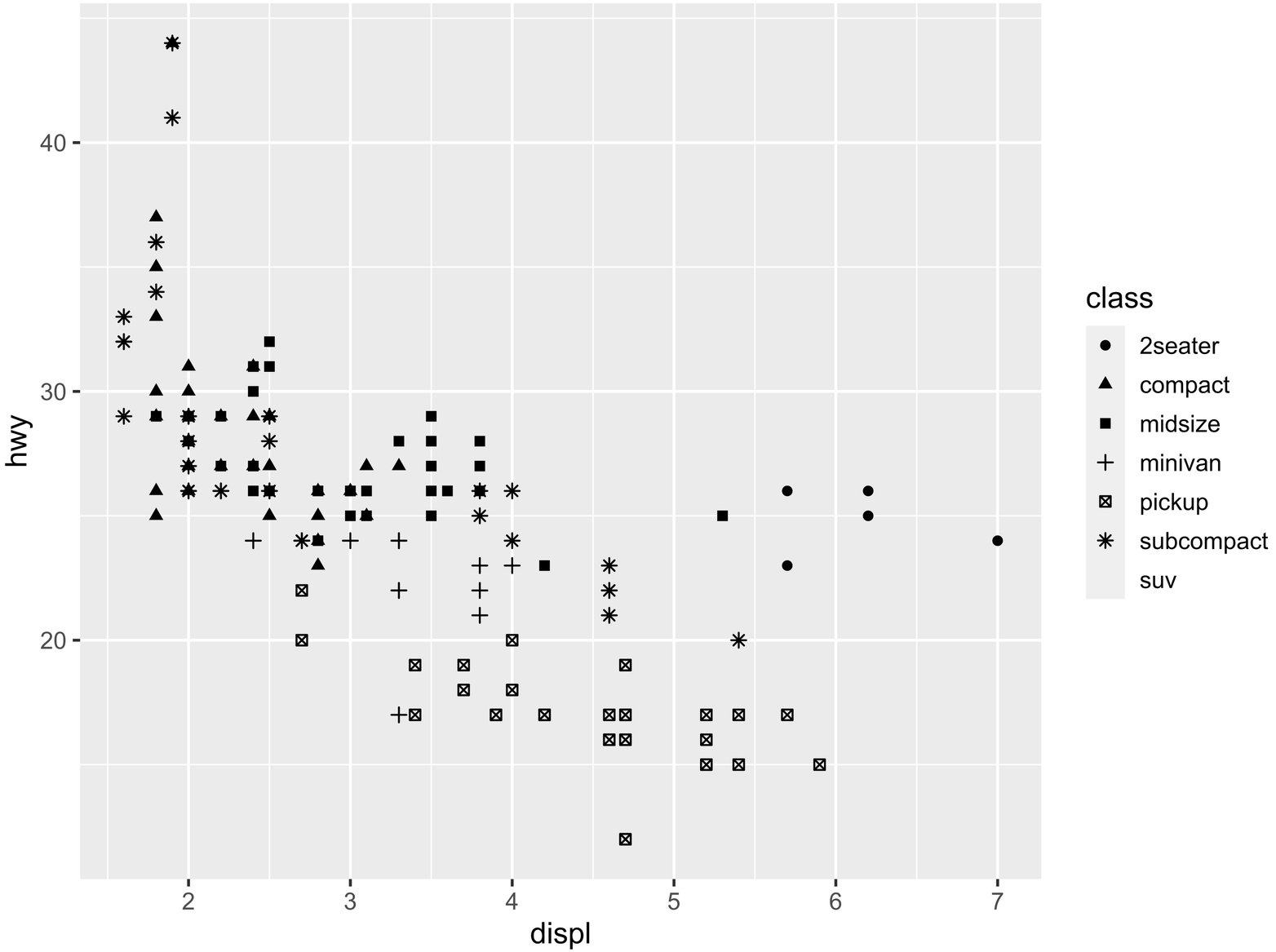
Atributo da variável classe

```
ggplot(mpg) +  geom_point( aes(x = displ, y = hwy, color = class) )
```



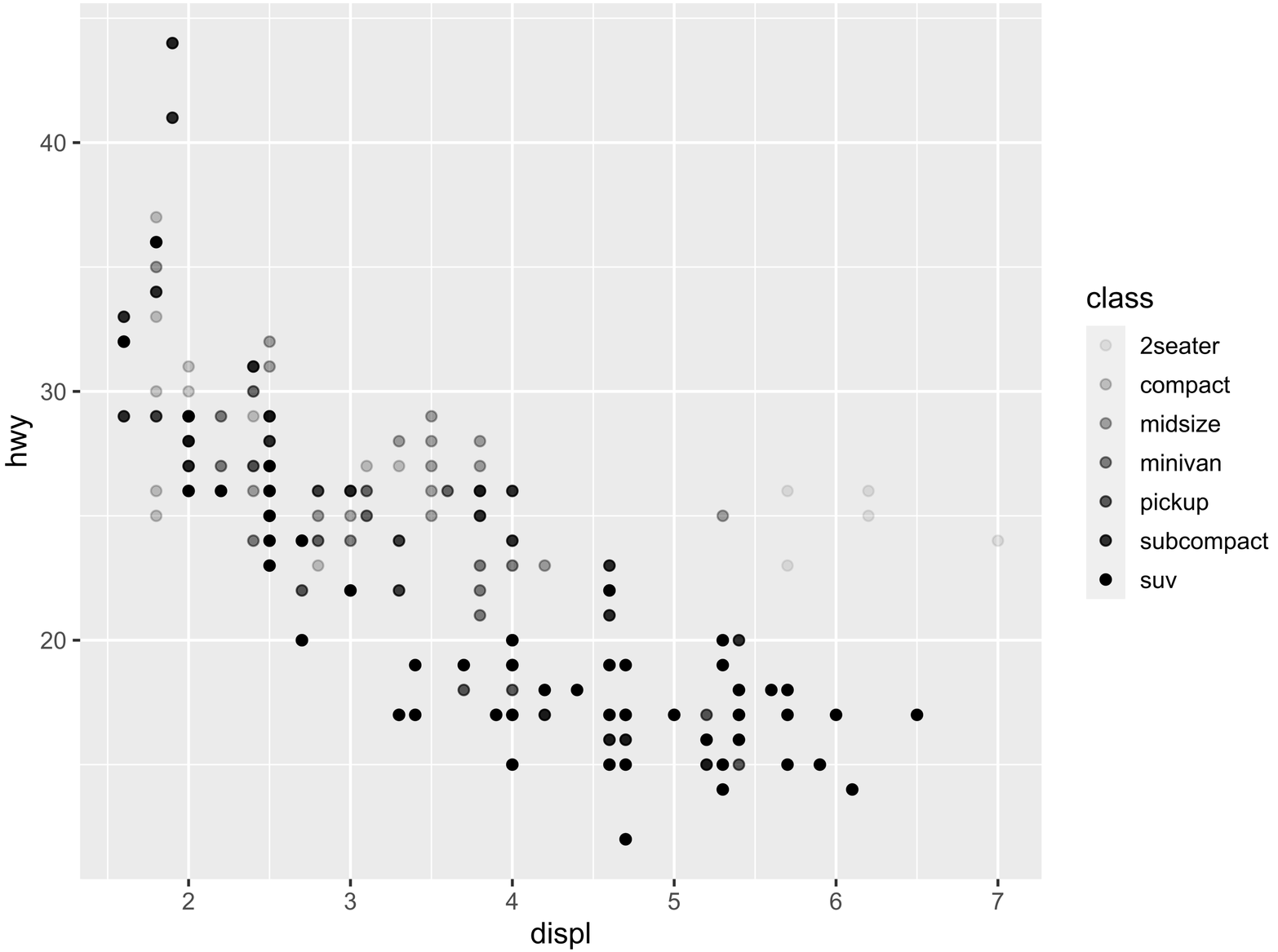
Atributo da variável classe

```
ggplot(mpg) +  geom_point( aes(x = displ, y = hwy, size = class) )
```



Atributo da variável classe

```
ggplot(mpg) +  geom_point( aes(x = displ, y = hwy, shape = class) )
```

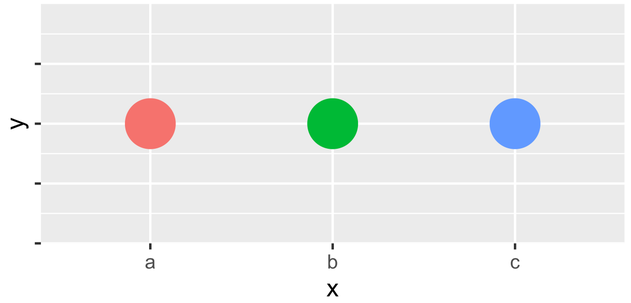
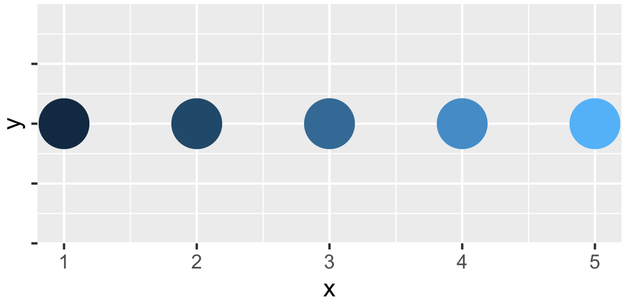
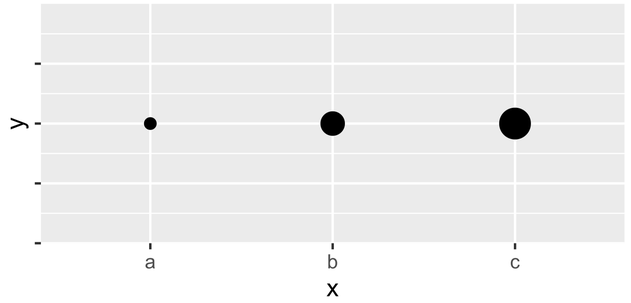
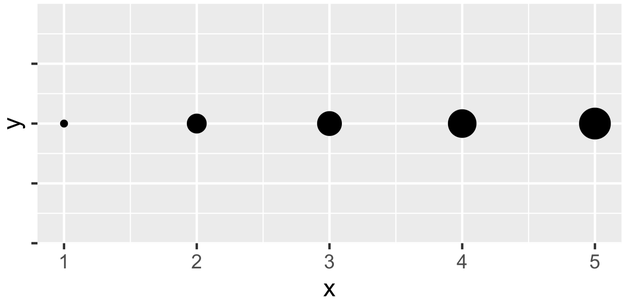
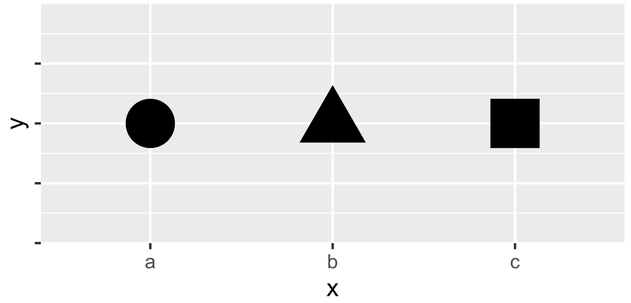


Atributo da variável classe

```
ggplot(mpg) +  geom_point( aes(x = displ, y = hwy, alpha = class) )
```

Sua vez

- Adicione cor, tamanho, forma e transparência aos seus gráficos.
- As coisas acontecem de forma diferente para variáveis discretas e contínuas?
- O que acontece quando você usa mais de uma estética?

Atributo	Discreto	Contínuo
color		
size		
shape		

Uma imagem não vale apenas mil palavras, é muito mais provável que seja examinada do que as palavras sejam lidas.

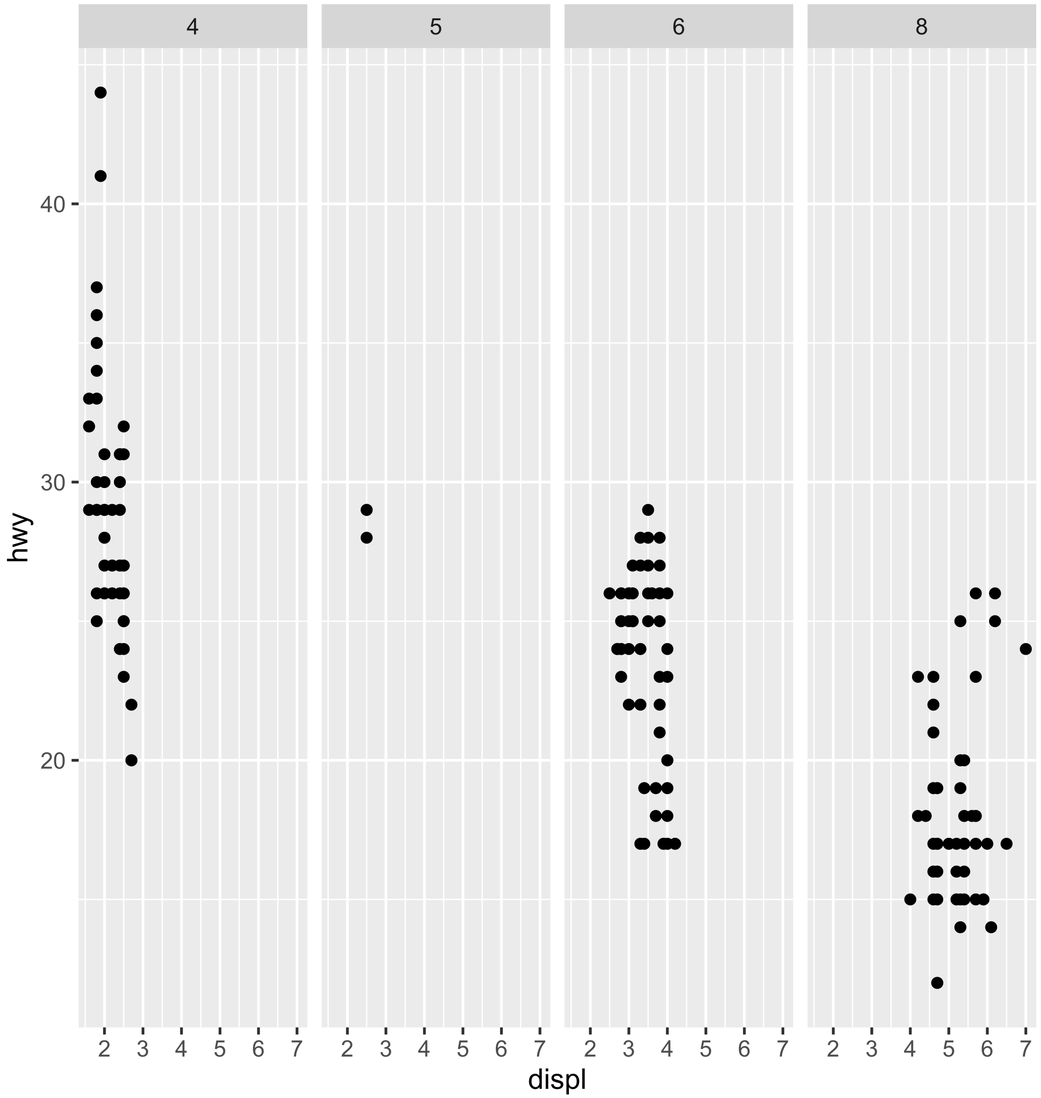
- John Tukey

Faceting

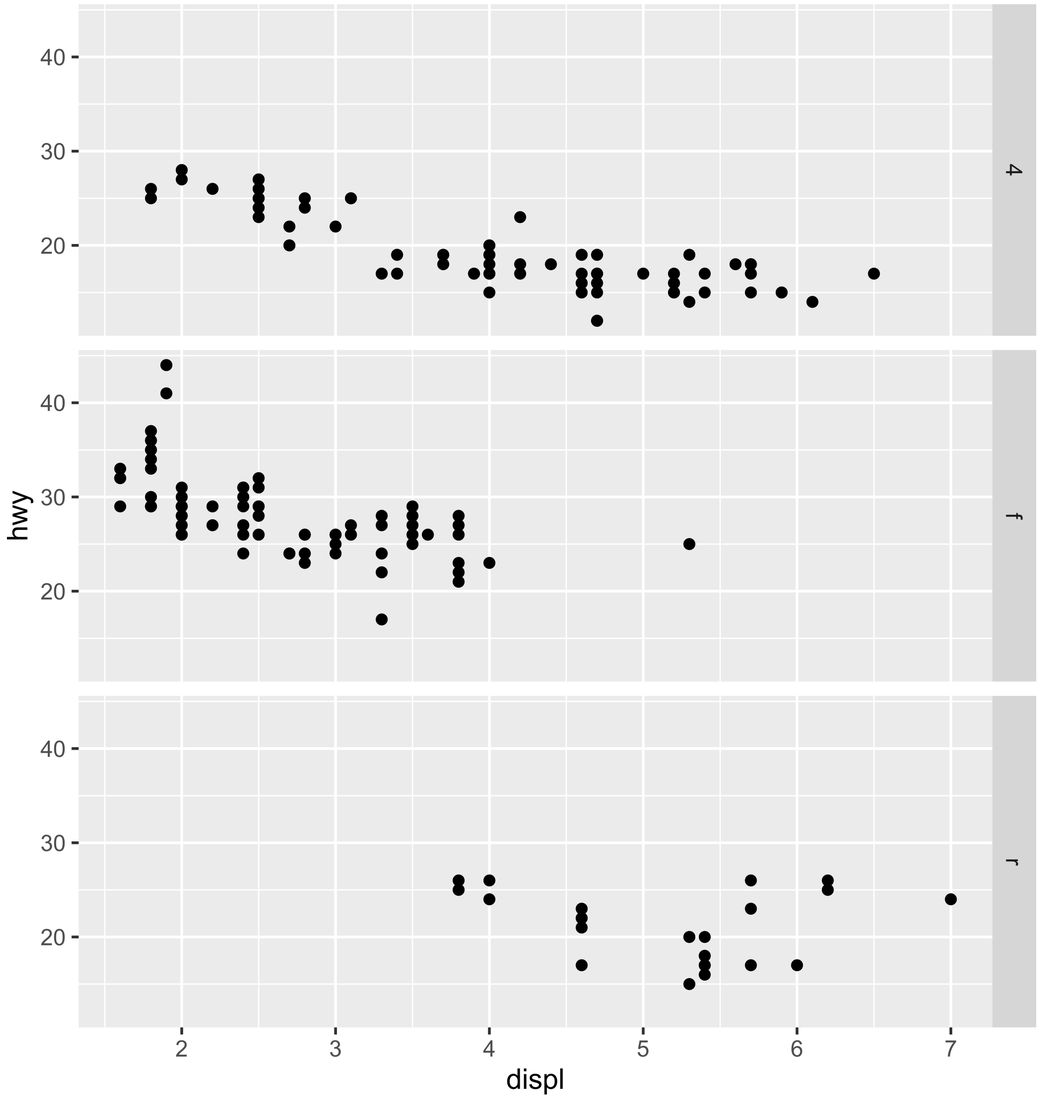
Subdivisões gráficas

Faceting

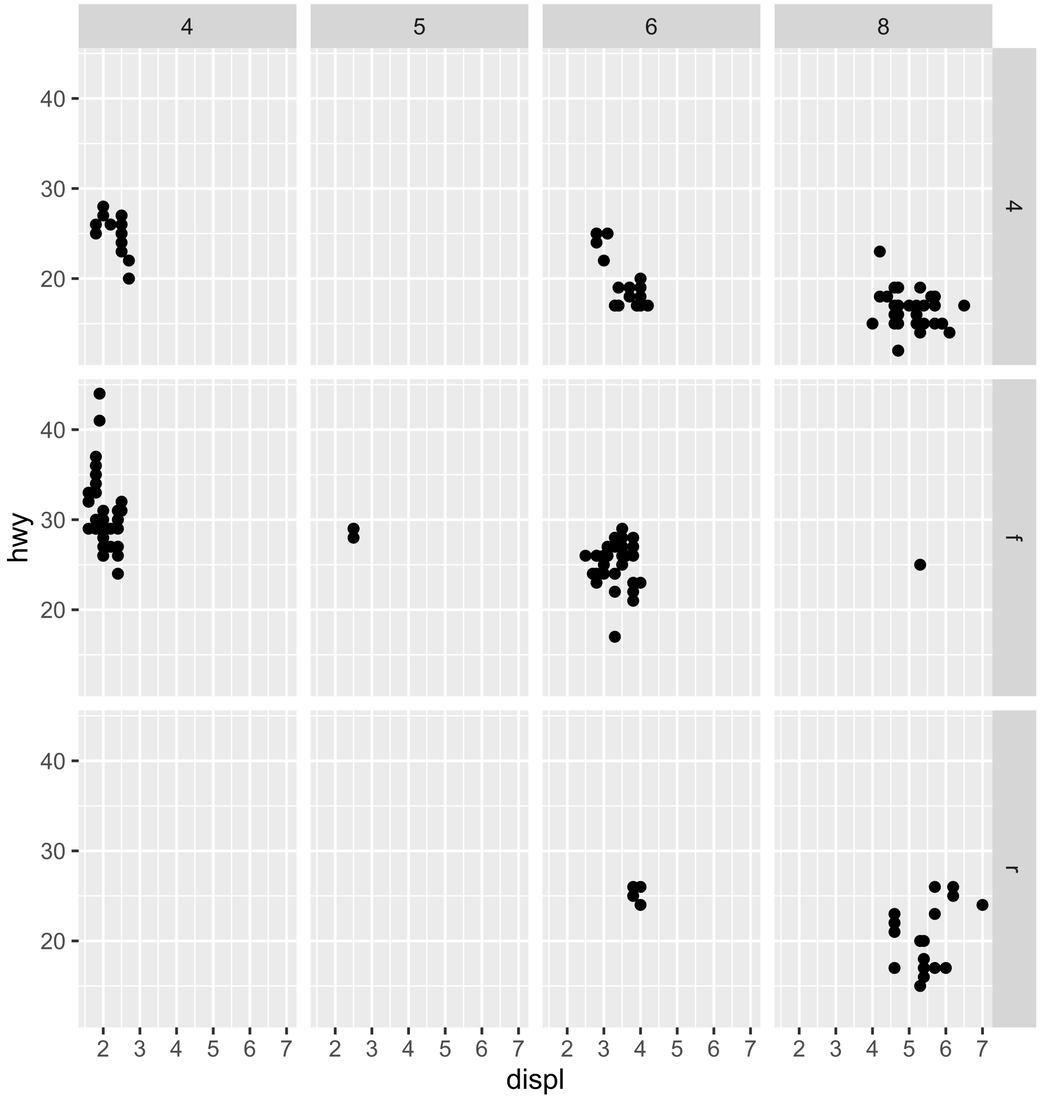
- Gráficos menores, porém “quebrados” em diferentes subconjuntos dos dados
- Útil para investigar aspectos condicionais nos dados
- Útil para grandes bases de dados



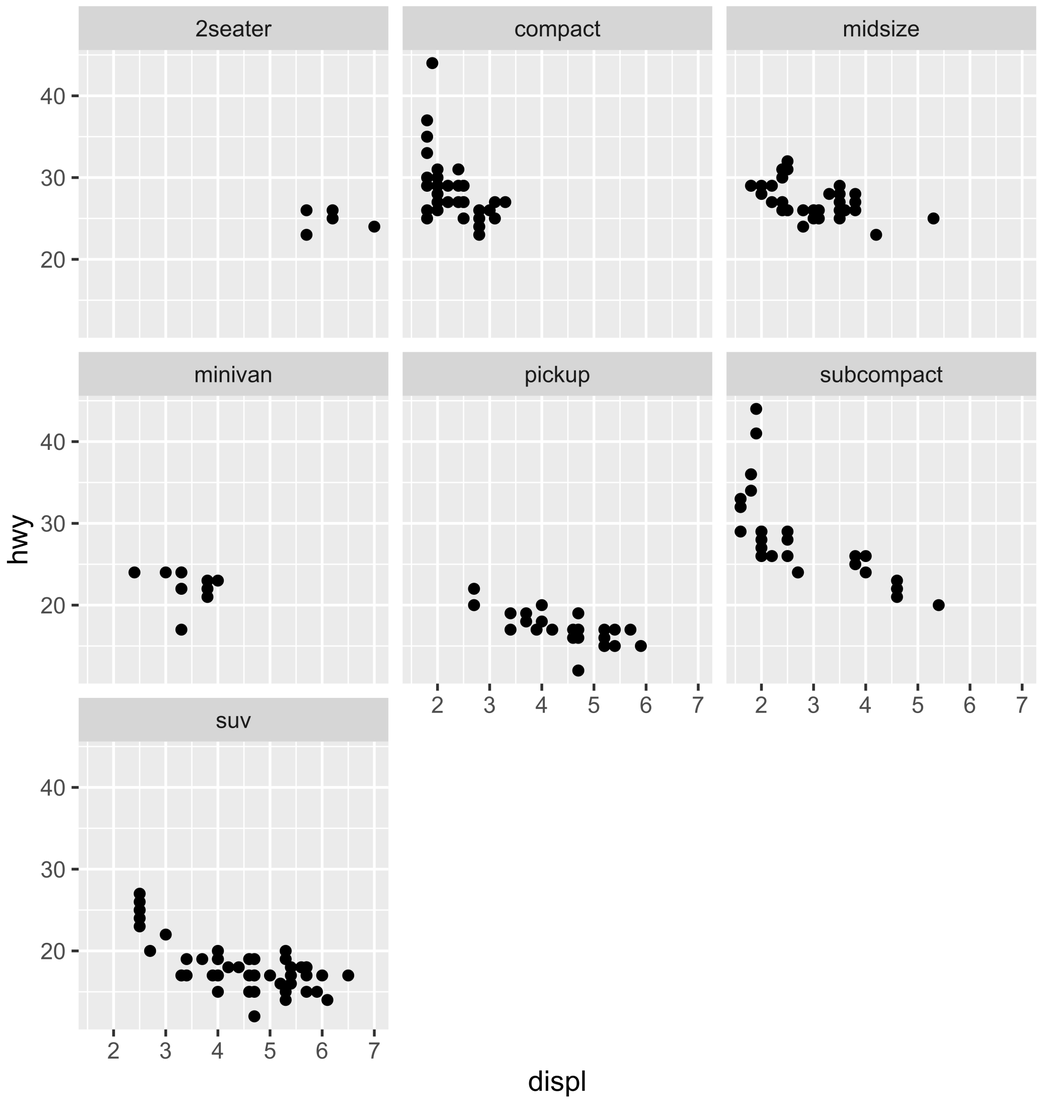

```
ggplot(mpg) +
 geom_point( aes(x = displ, y = hwy) ) +
 facet_grid( . ~ cyl )
```




```
ggplot(mpg) +
 geom_point( aes(x = displ, y = hwy) ) +
 facet_grid( drv ~ . )
```




```
ggplot(mpg) +
  geom_point( aes(x = displ, y = hwy) ) +
  facet_grid( drv ~ cyl )
```




```
ggplot(mpg) +
  geom_point( aes(x = displ, y = hwy) ) +
  facet_wrap( ~ class )
```

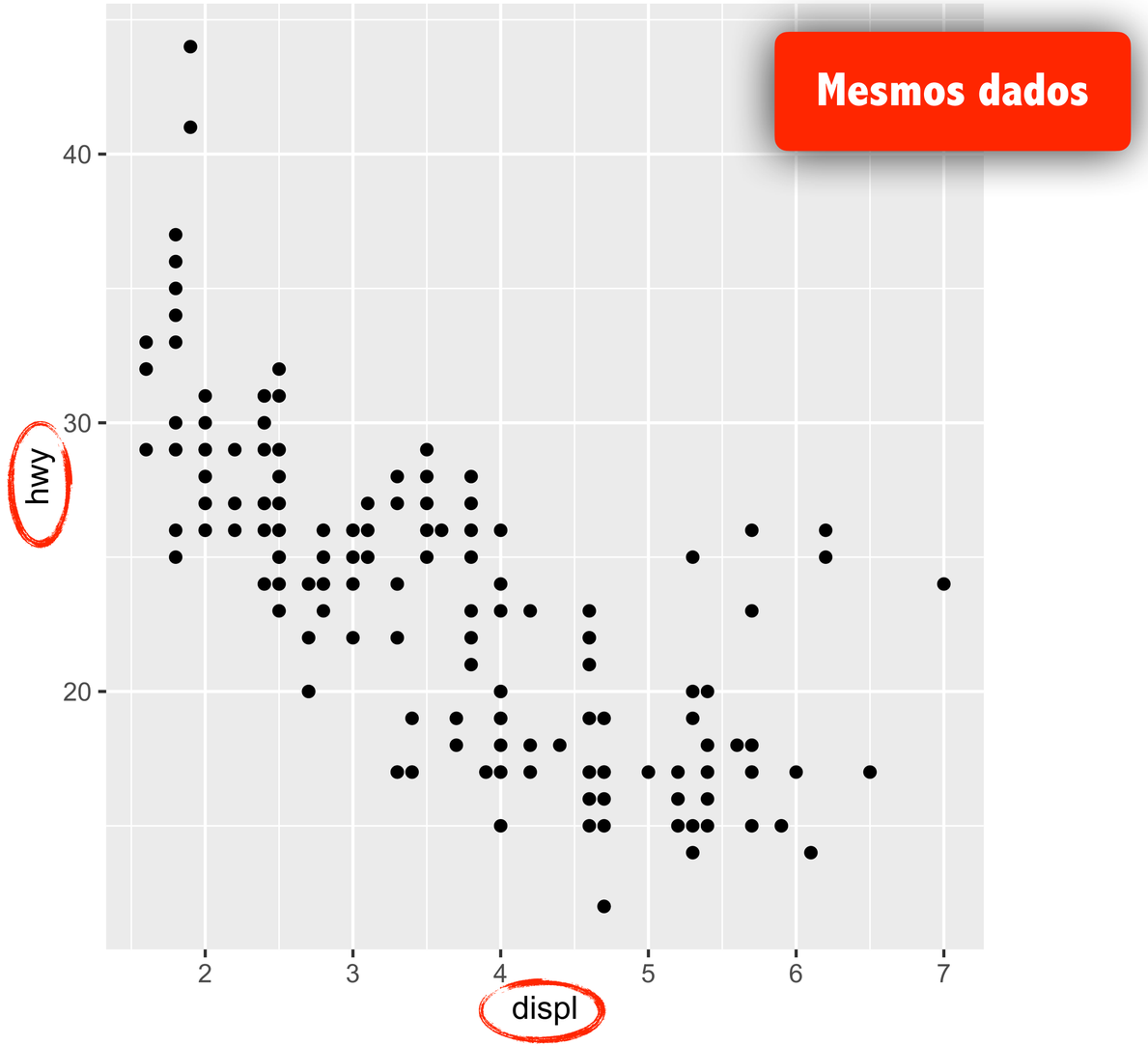
Resumo

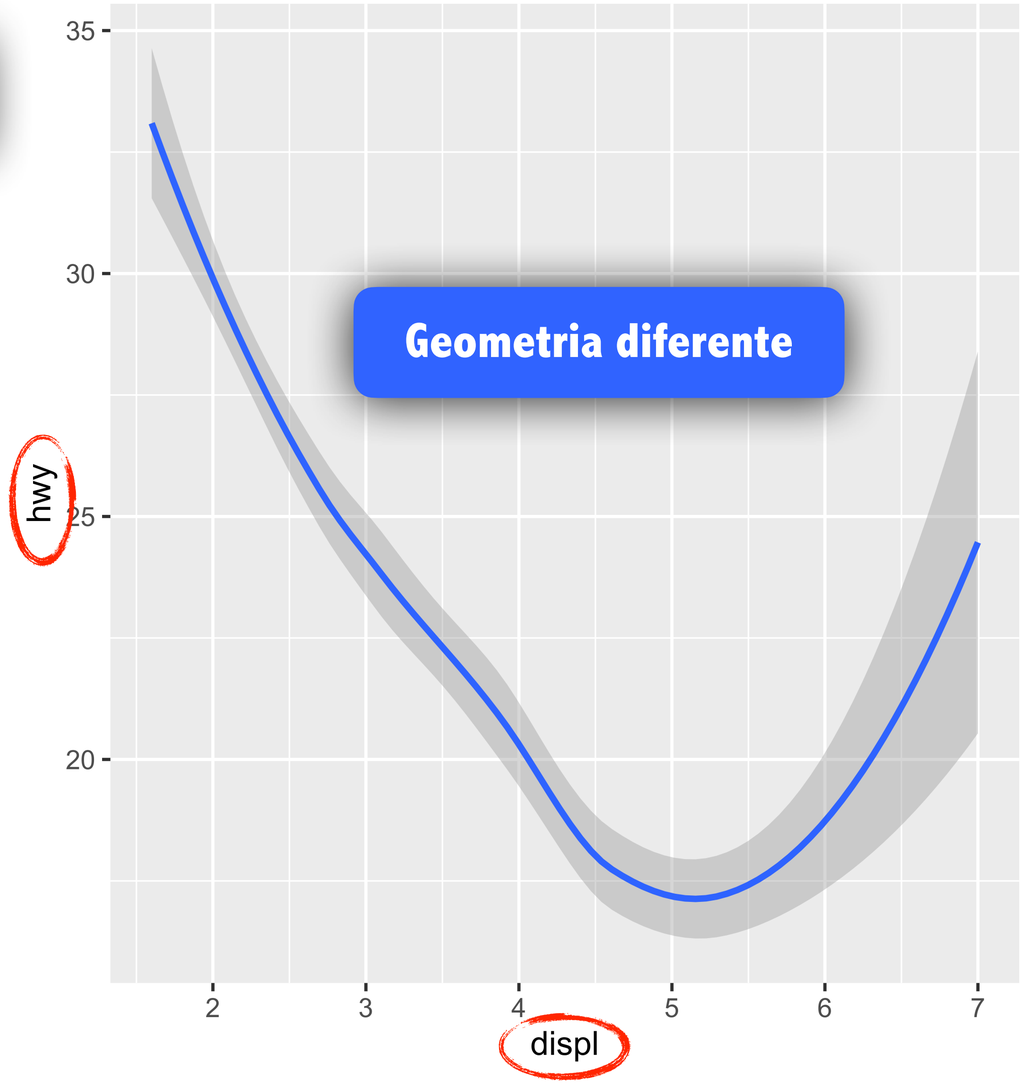
`facet_grid( )` : categorias alinhadas, por níveis, em uma mesma dimensão

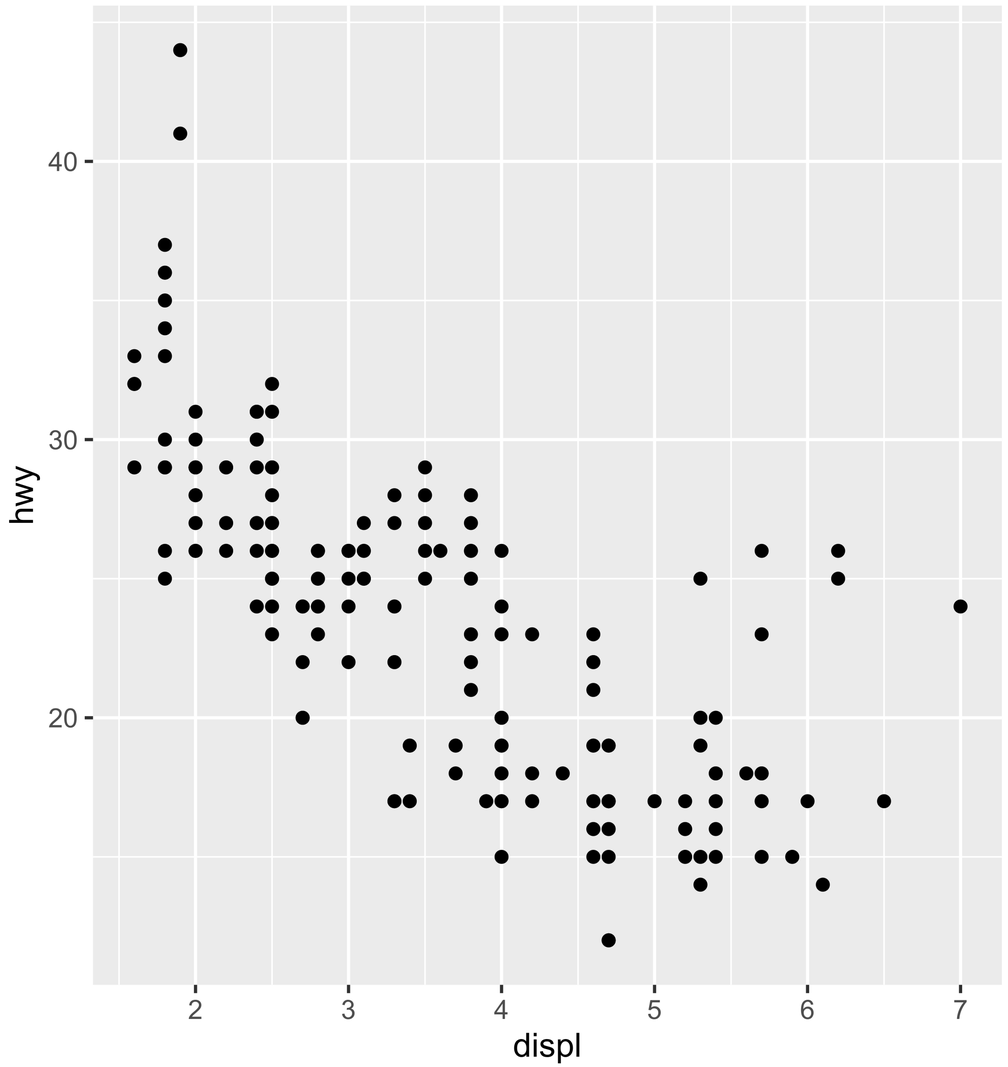
`facet_wrap( )` : categorias em sequência, por níveis, em mais de uma dimensão

Suavização

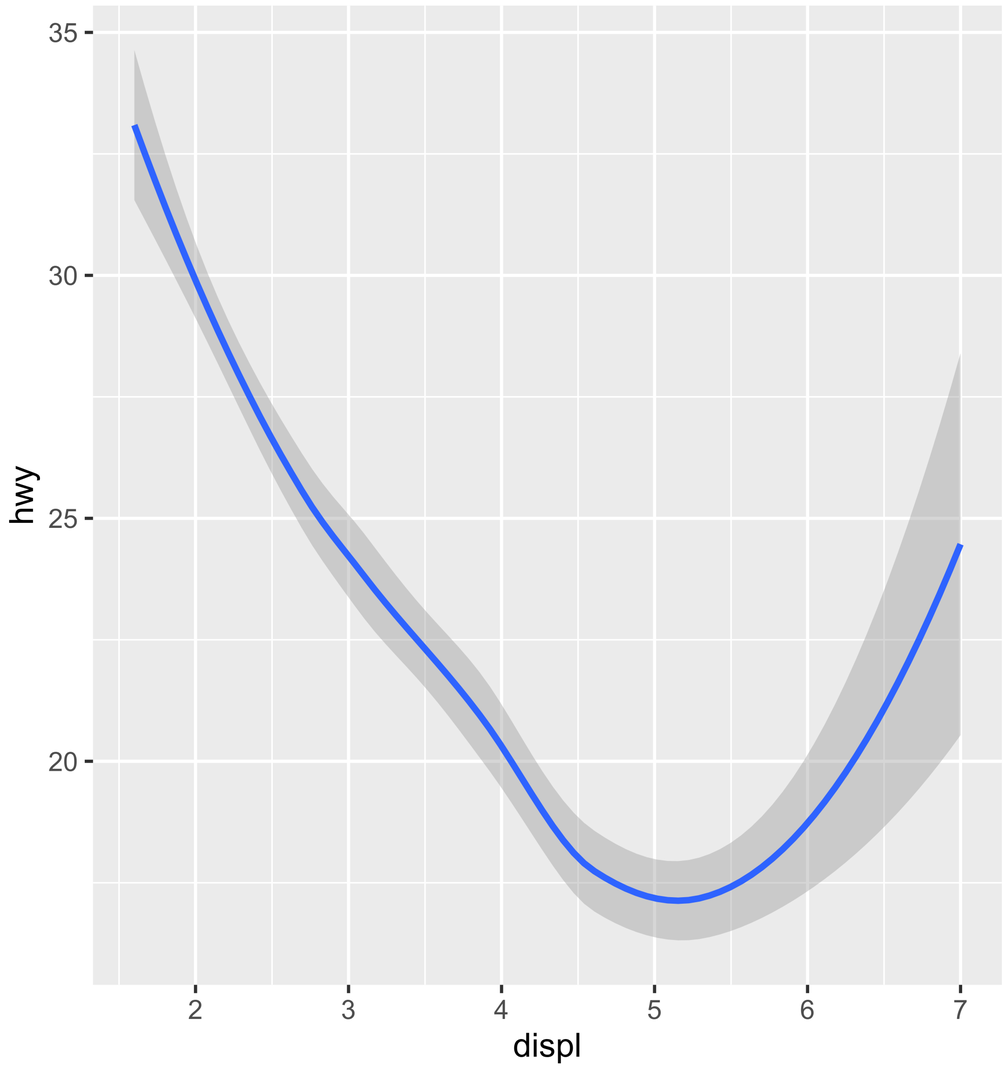
Há similaridades entre esses gráficos?





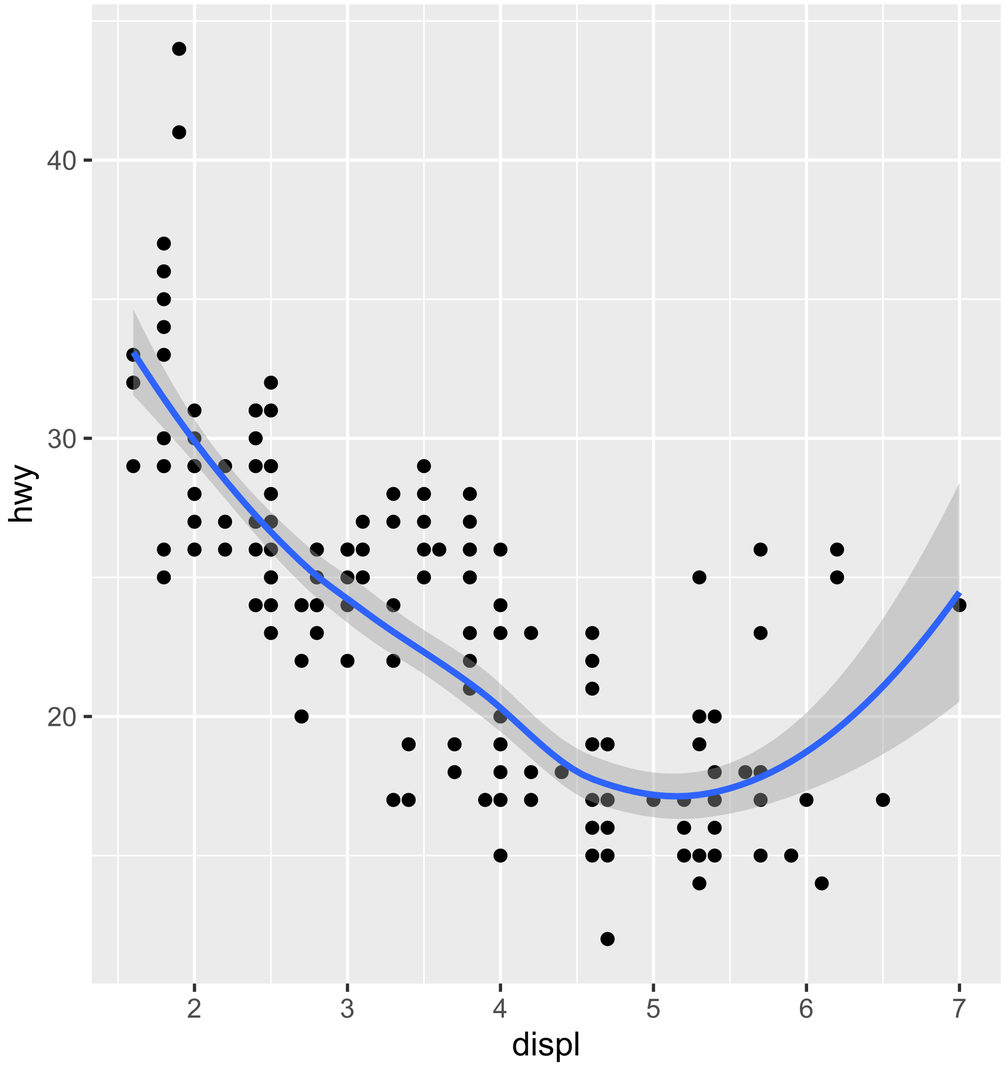



```
ggplot(mpg) +
  geom_point( aes(x = displ, y = hwy) )
```




```
ggplot(mpg) +
  geom_smooth( aes(x = displ, y = hwy) )
```

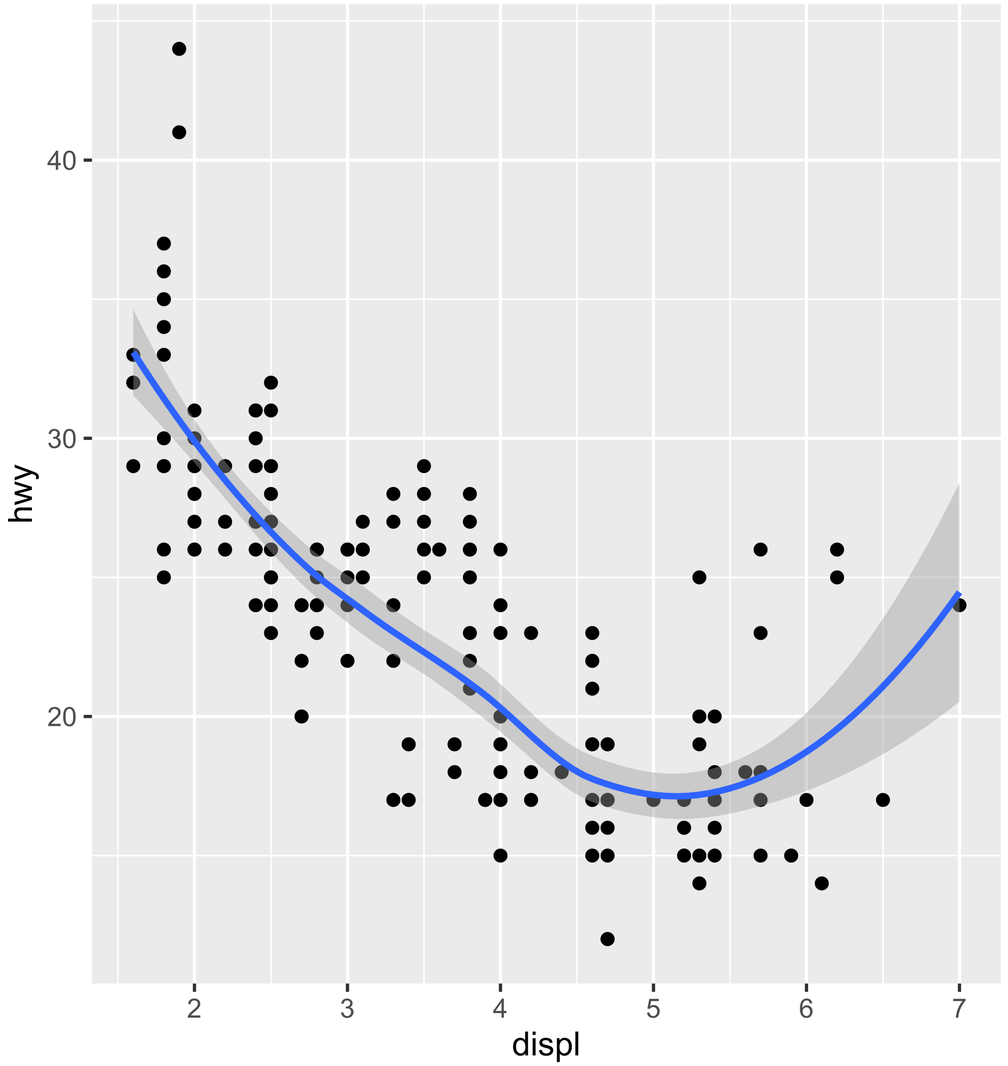
Podemos adicionar mais de uma geometria?




```
ggplot(mpg) +
 geom_point( aes(x = displ, y = hwy) ) +
 geom_smooth( aes(x = displ, y = hwy) )
```

Local

Podemos adicionar mais de uma geometria?



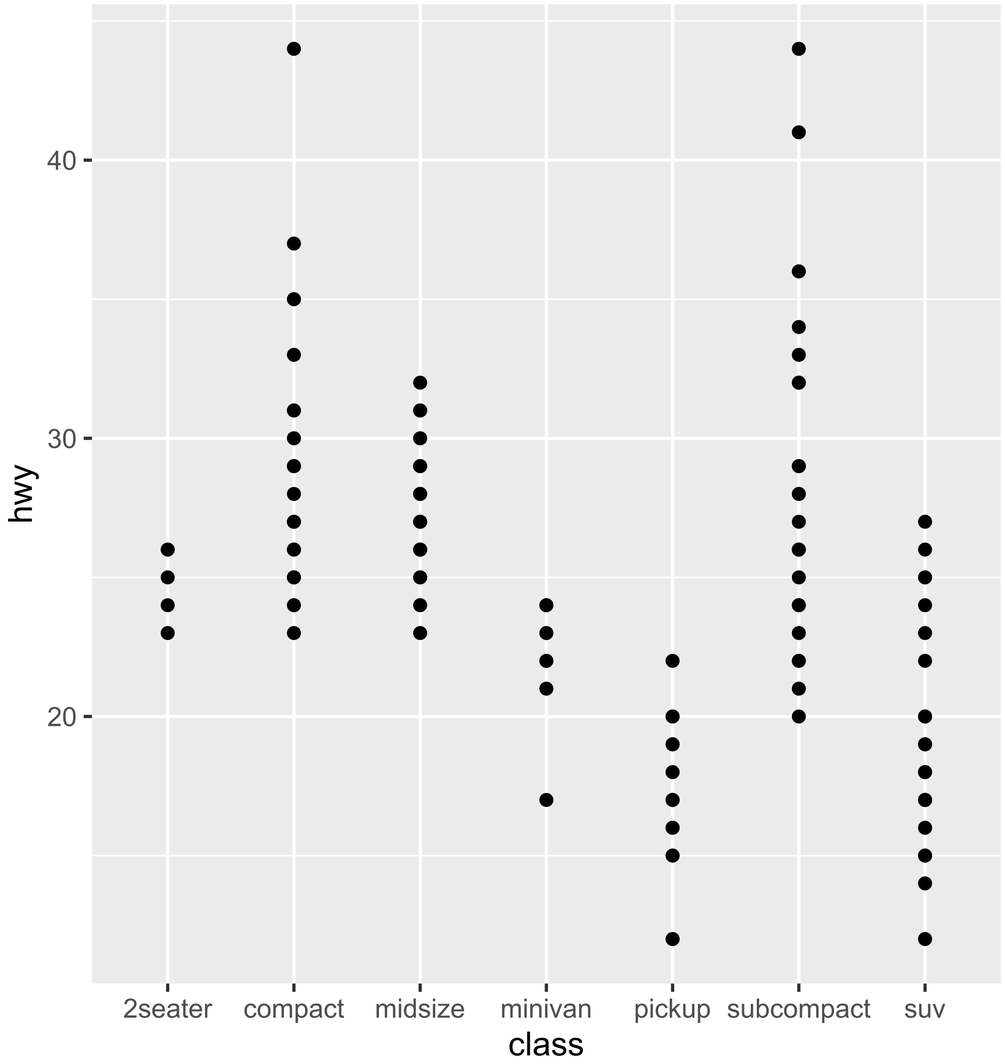

```
ggplot(mpg, aes(x = displ, y = hwy) ) +
  geom_point() +
  geom_smooth()
```

Global

Boxplot

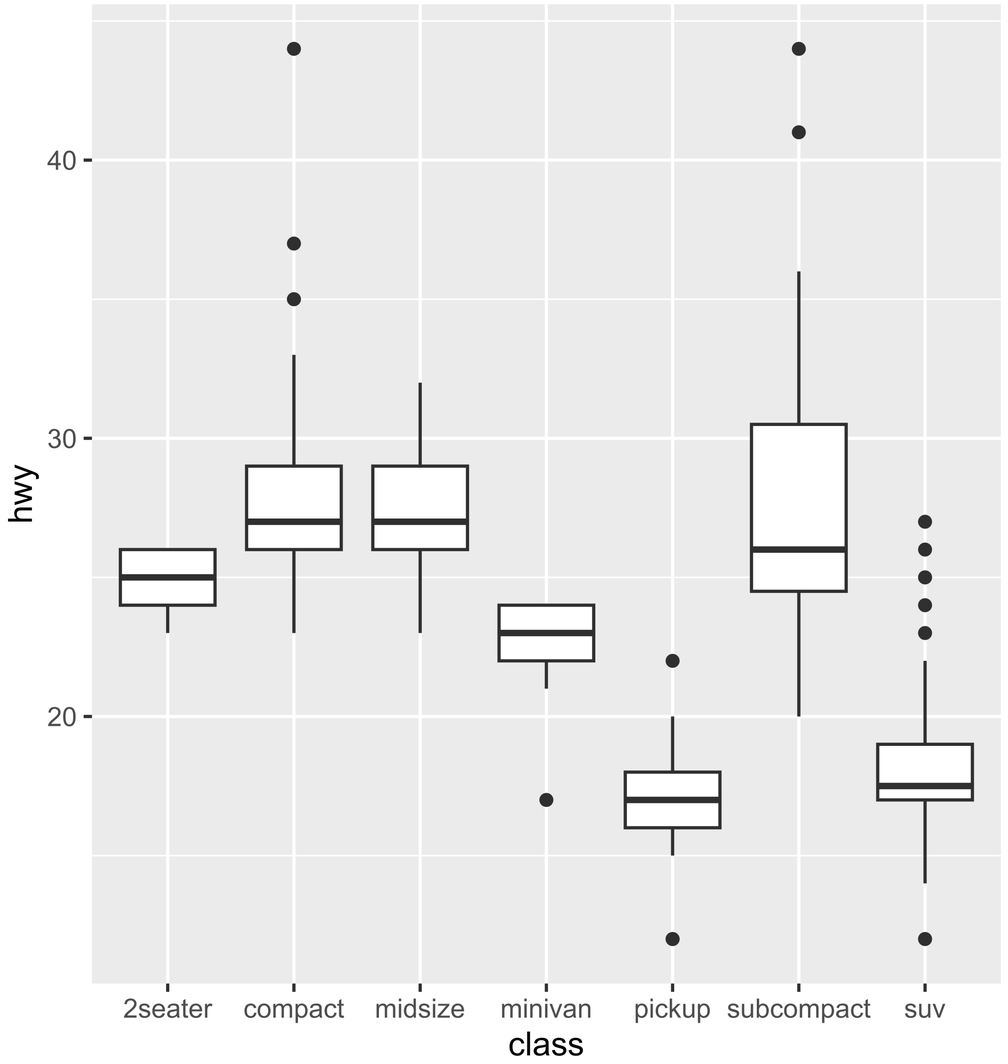
2 dimensões: categórico x contínuo

Esse gráfico está bom? Que melhoria você sugere?



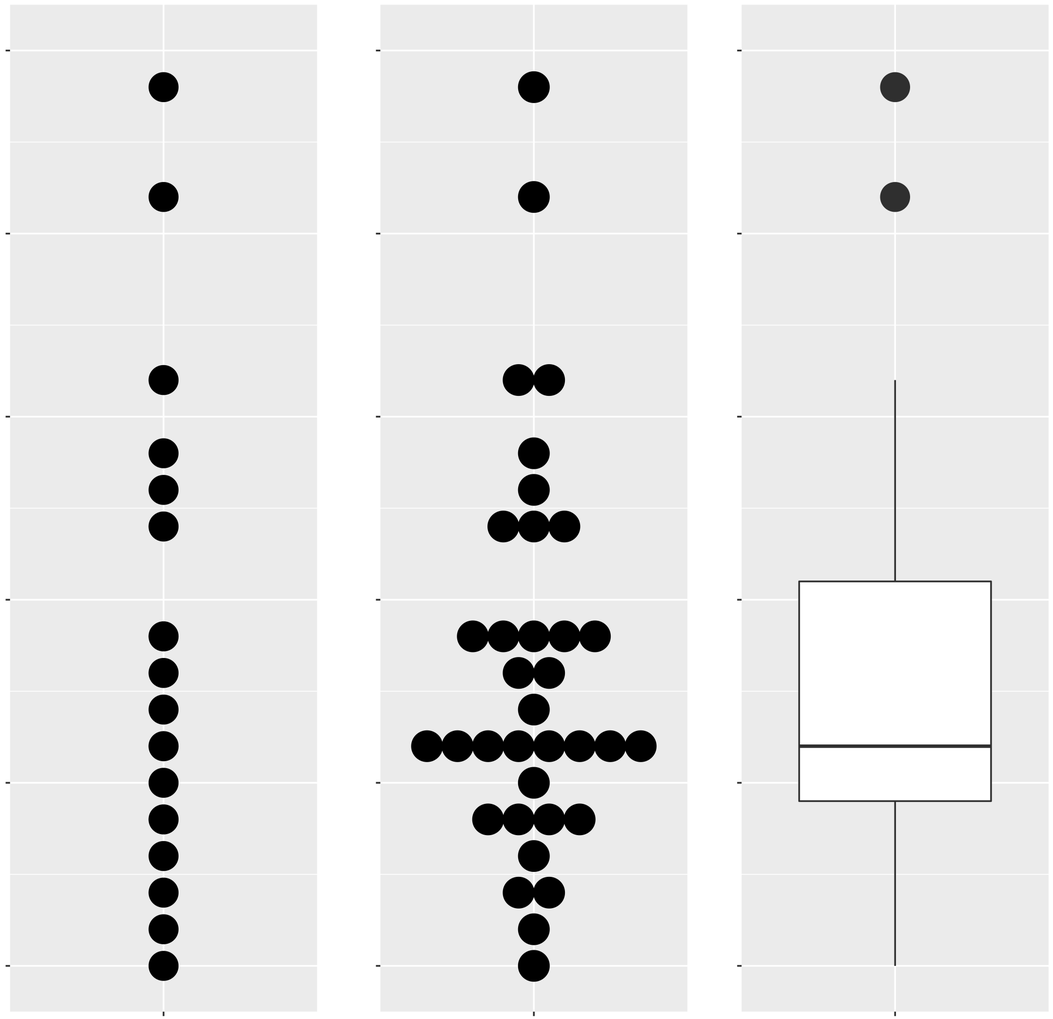

```
ggplot( mpg, aes(x = class, y = hwy) ) +
 geom_point()
```

Esse gráfico está bom? Que melhoria você sugere?

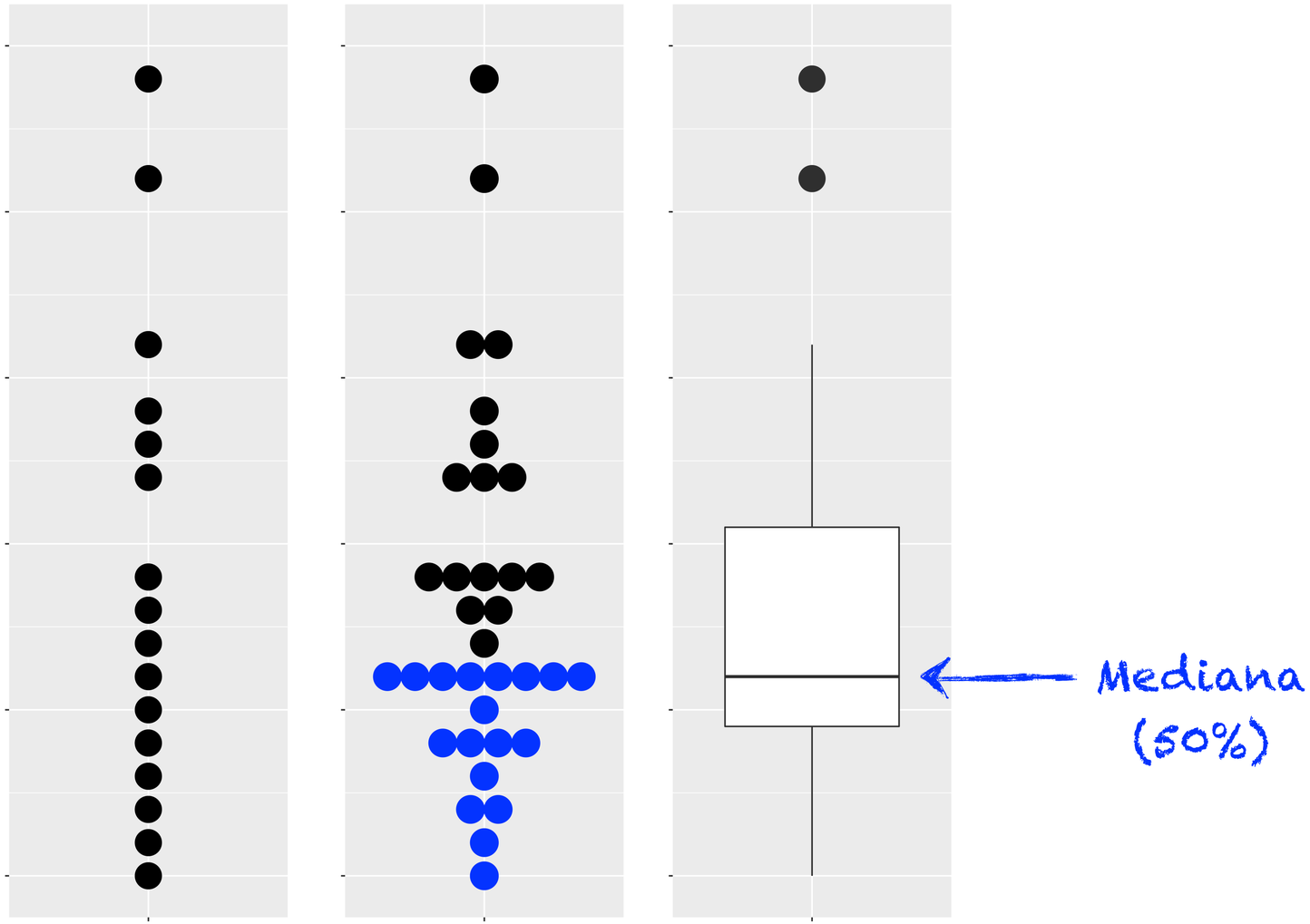



```
ggplot( mpg, aes(x = class, y = hwy) ) +
 geom_boxplot()
```

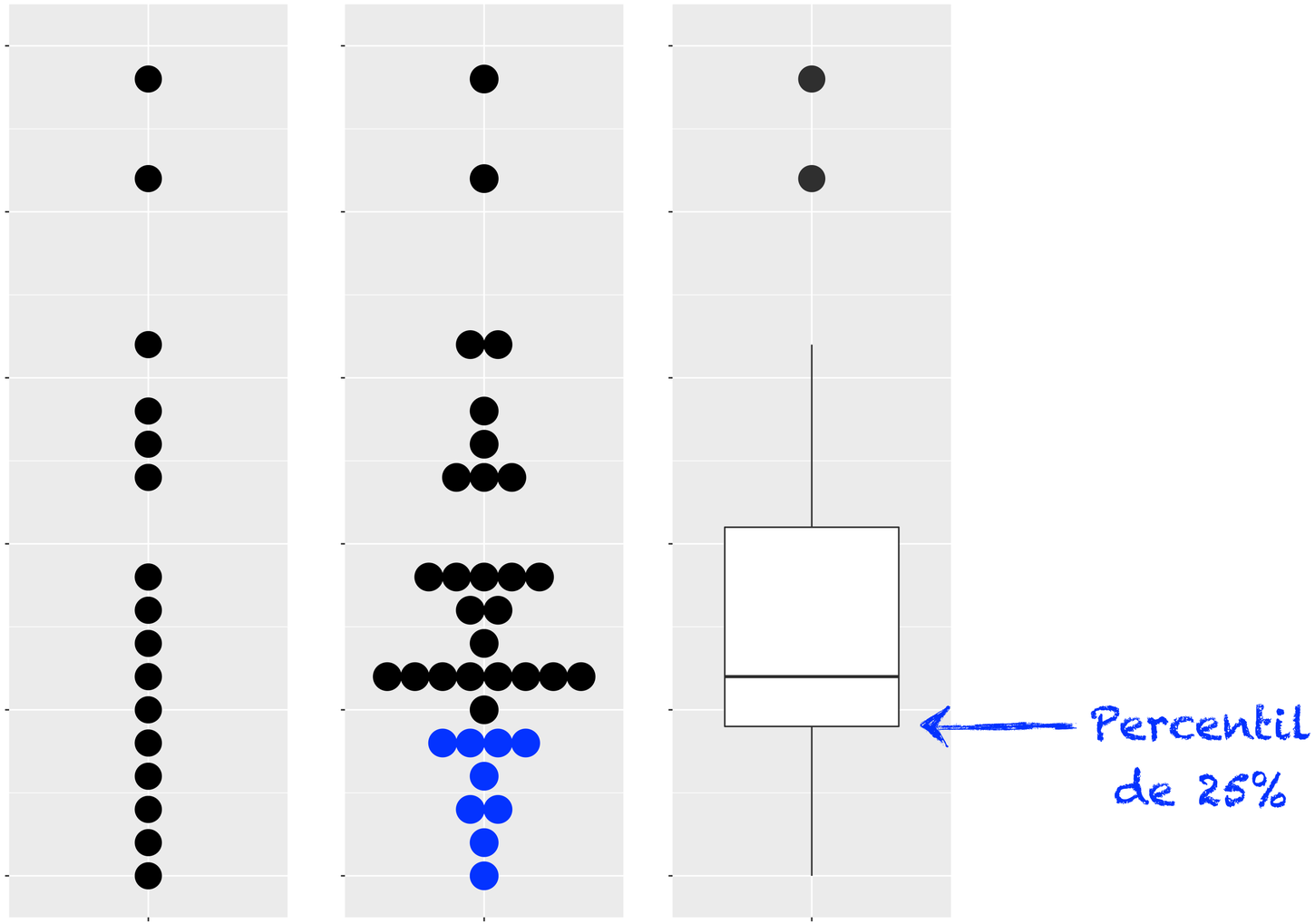
Boxplots



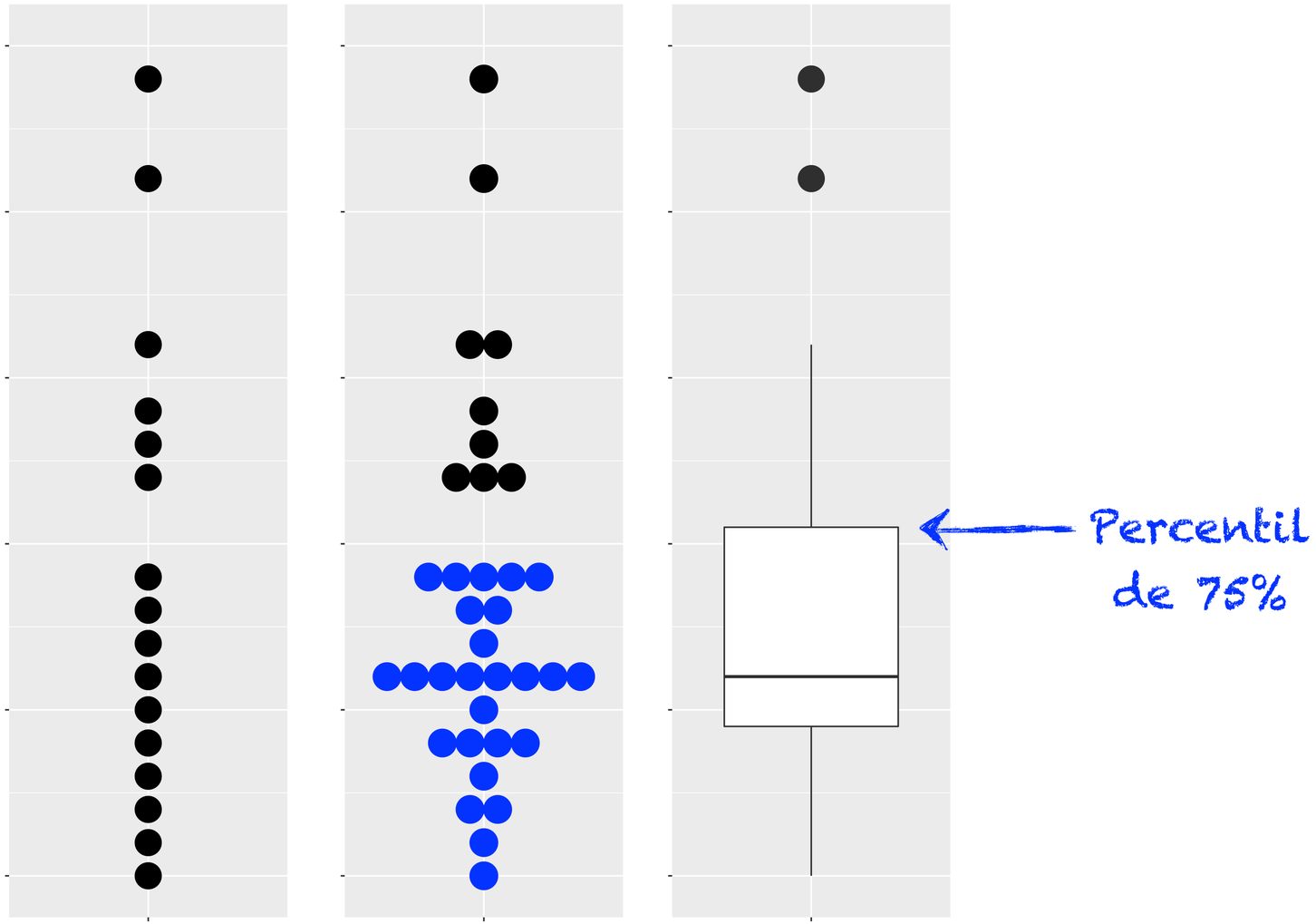
Boxplots



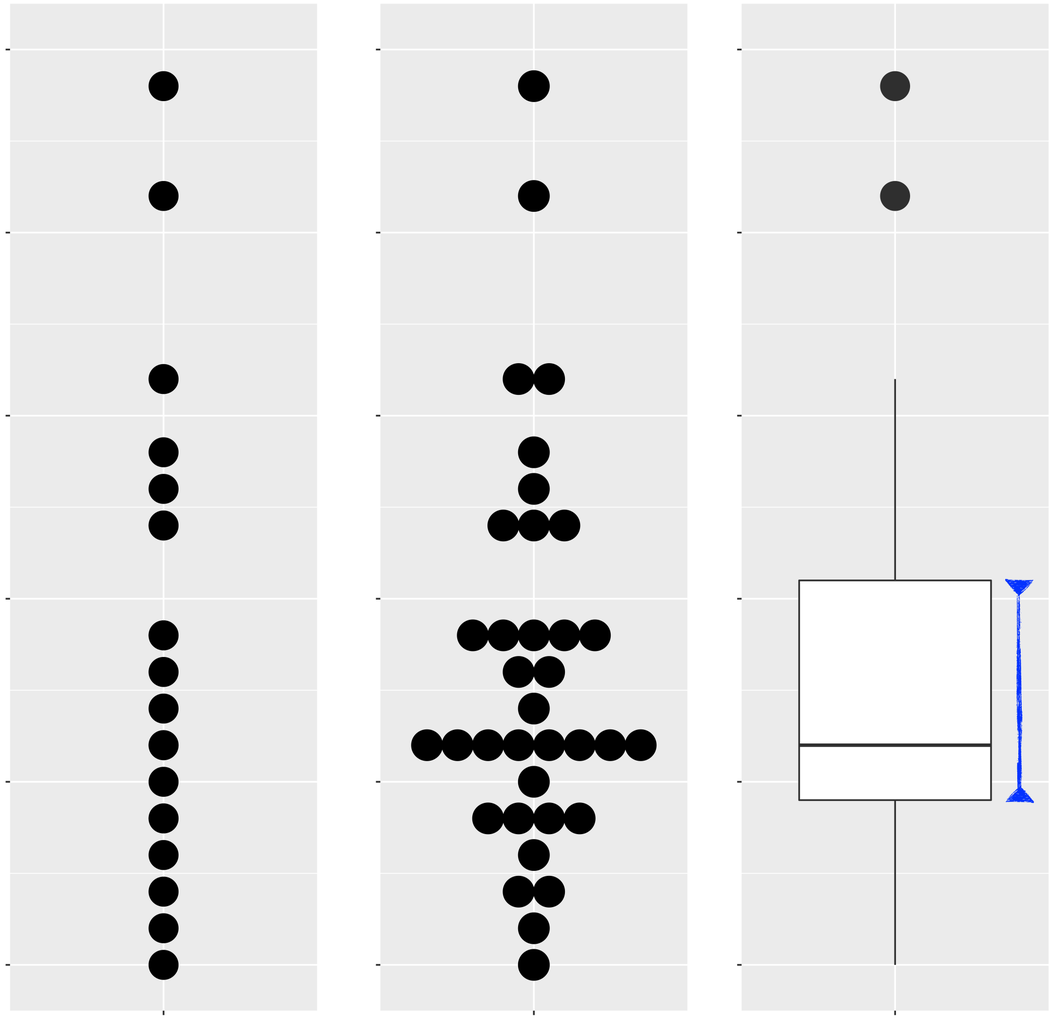
Boxplots



Boxplots

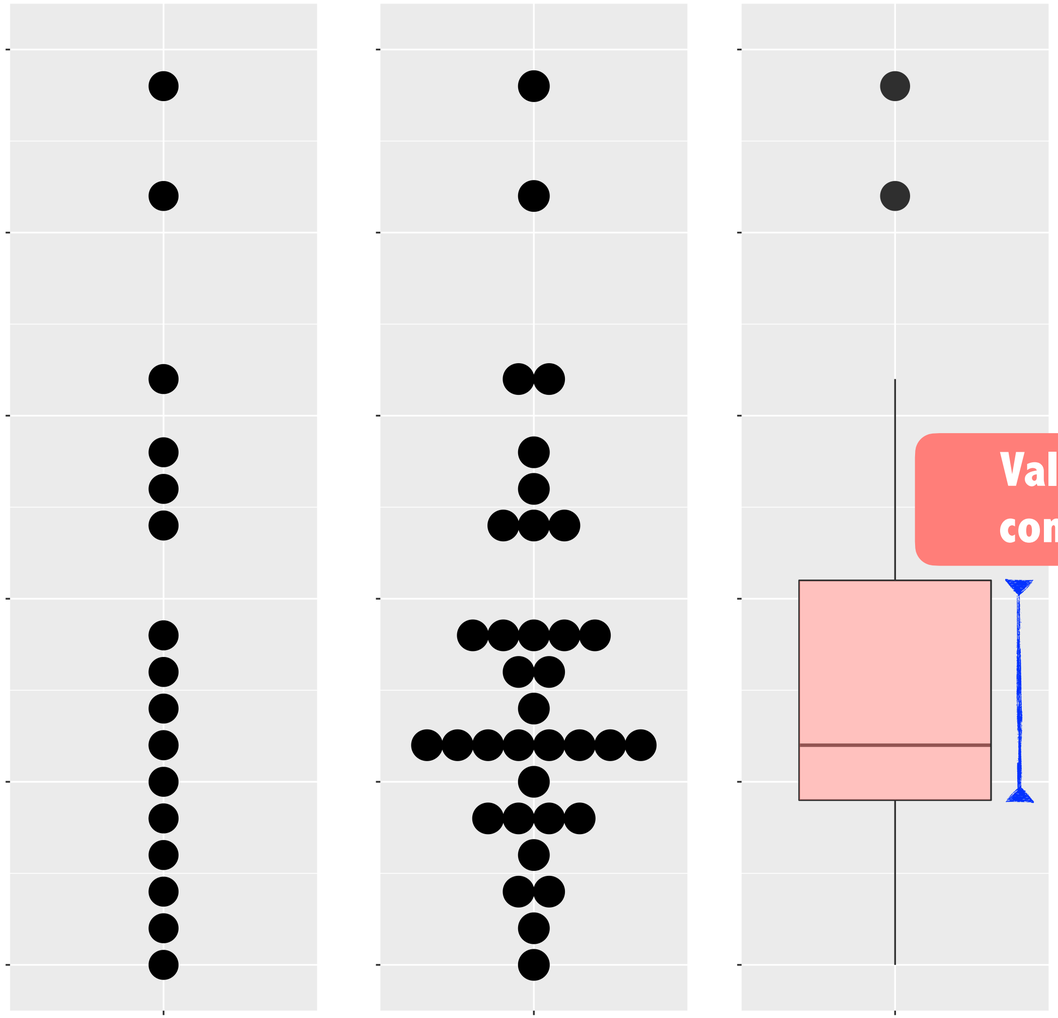


Boxplots



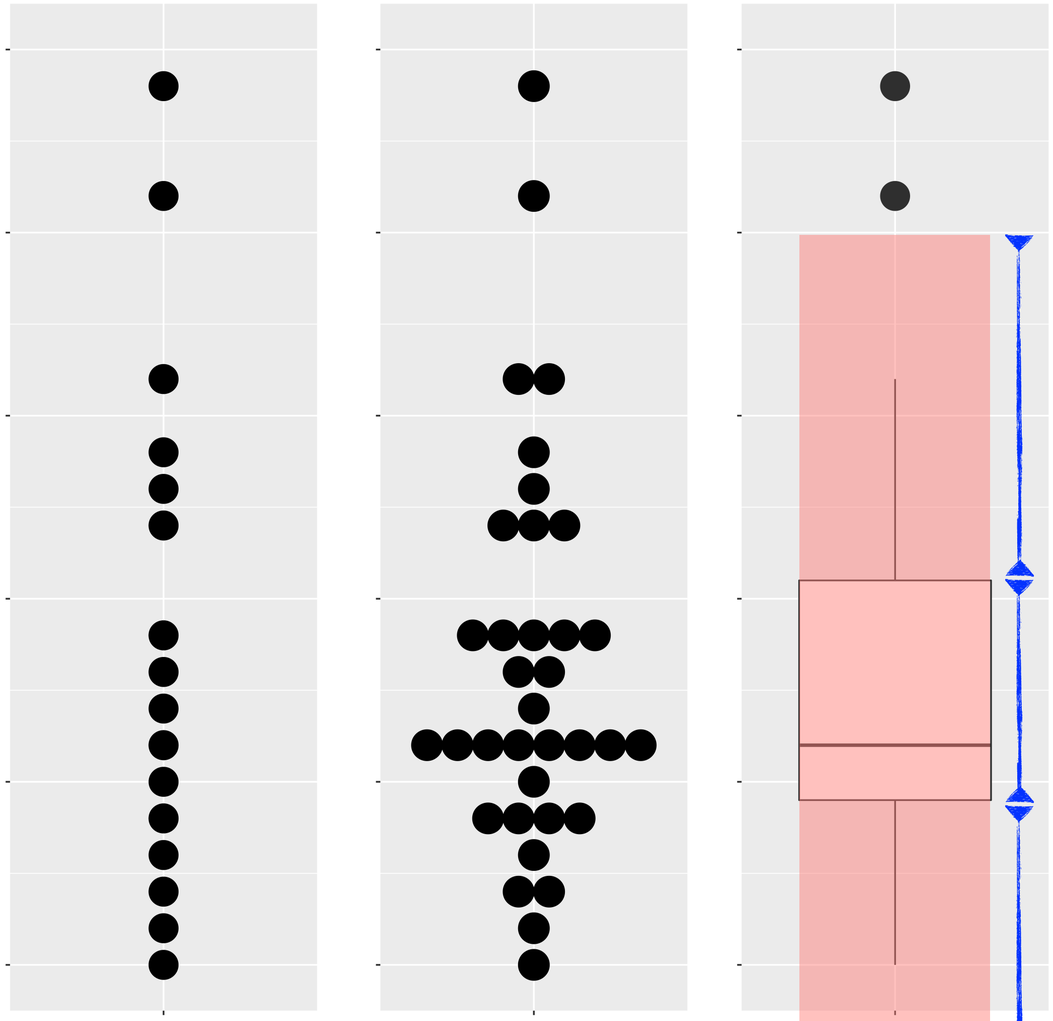
IIQ = Percentil de 75% - Percentil de 25%

Boxplots



IIQ = Percentil de 75% - Percentil de 25%

Boxplots



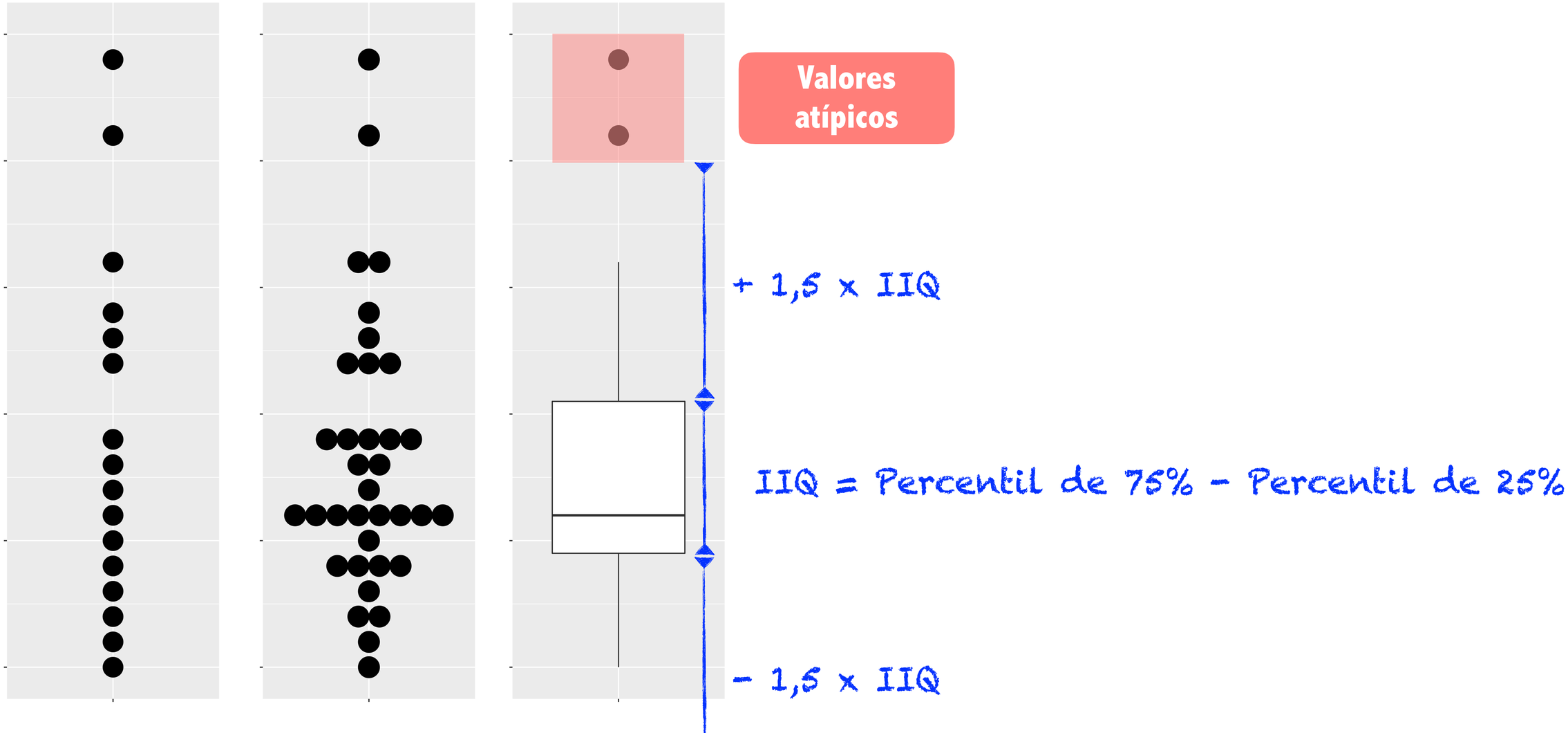
Valores típicos

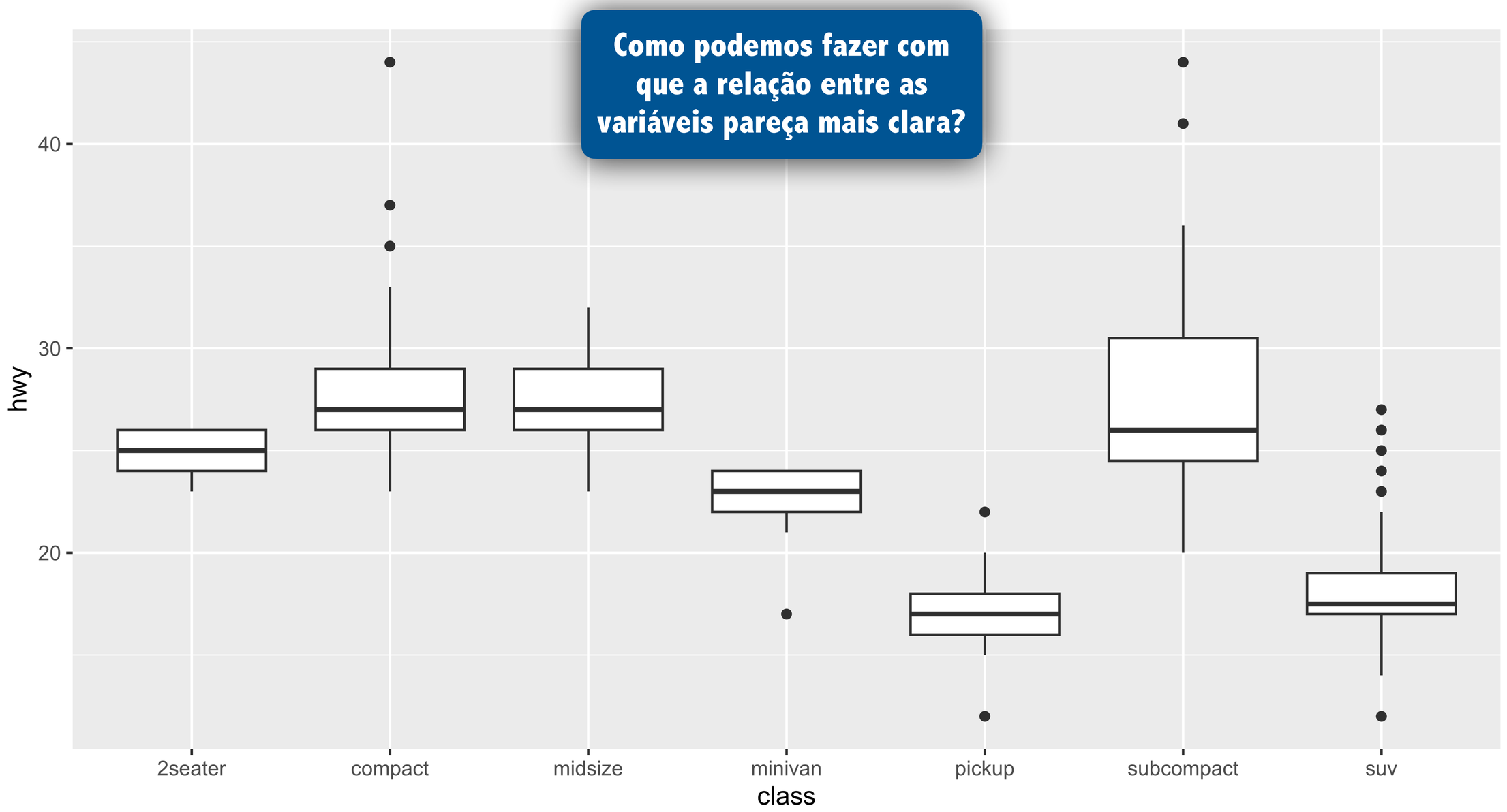
+ 1,5 x IIQ

IIQ = Percentil de 75% - Percentil de 25%

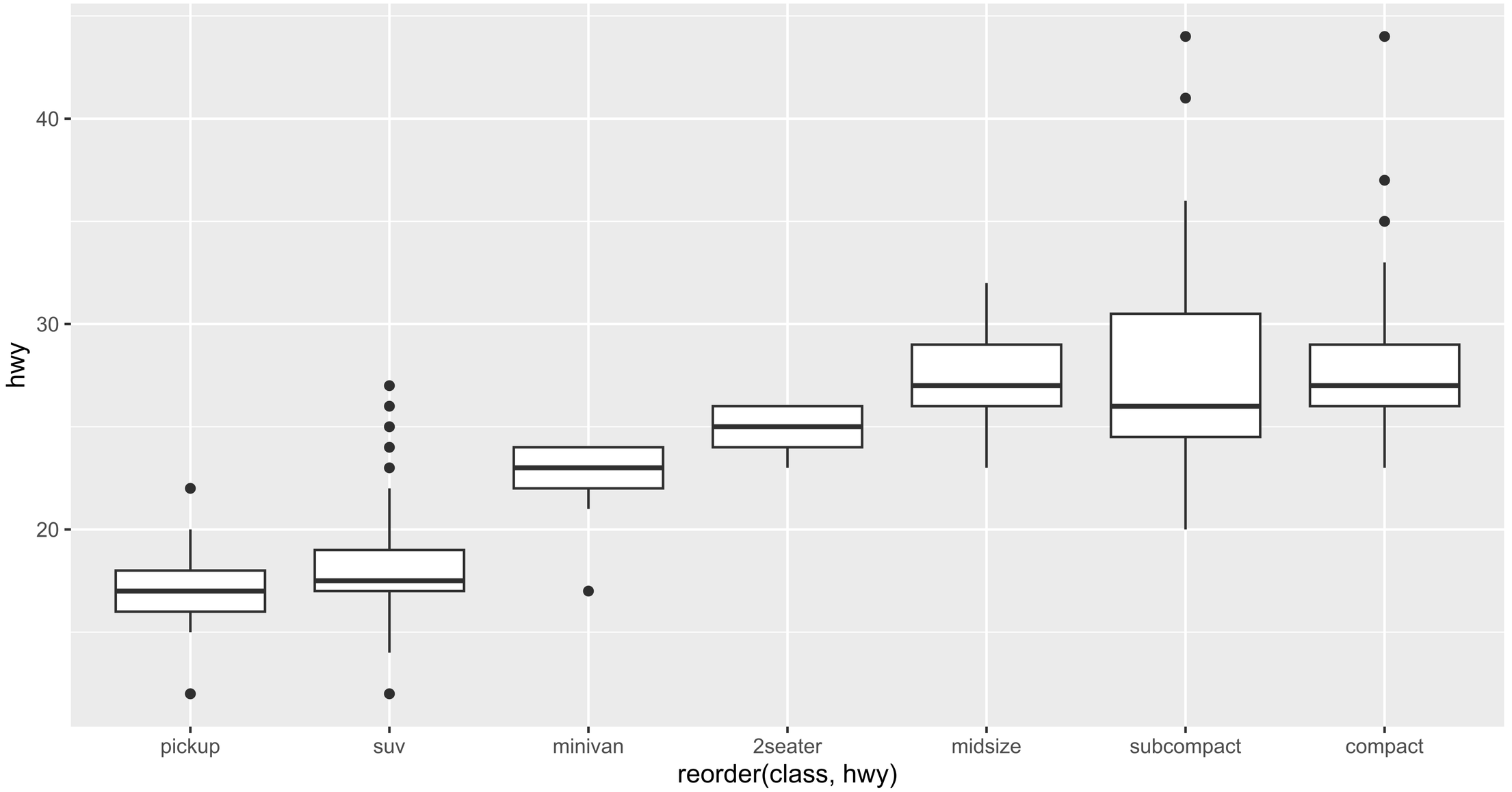
- 1,5 x IIQ

Boxplots


Valores atípicos
+ 1,5 x IIQ
IIQ = Percentil de 75% - Percentil de 25%
- 1,5 x IIQ




```
ggplot( mpg, aes(x = class, y = hwy) ) + geom_boxplot()
```




```
ggplot( mpg, aes(x = reorder(class, hwy), y = hwy) ) + geom_boxplot()
```

Sua vez

- O comando `reorder`, ordena em função da média
- Adapte o código para que a ordenação seja por mediana

  ★ Utilize o help

  ✓ Procure informações relevantes

  ✓ Ignore o que não fizer sentido

  ✓ Tente os exemplos

Description

Útil visão geral

Usage

Ótimo local para visualizar valores padrões

Arguments

Explicação de cada argumento

Value

O que a função retorna

Examples

Seção mais útil entre todas!

reorder.default {stats} R Documentation

Reorder Levels of a Factor

Description

`reorder` is a generic function. The `"default"` method treats its first argument as a categorical variable, and reorders its levels based on the values of a second variable, usually numeric.

Usage

```
reorder(x, ...)

## Default S3 method:
reorder(x, X, FUN = mean, ...,
        order = is.ordered(x), decreasing = FALSE)
```

Arguments

`x`	an atomic vector, usually a `factor` (possibly ordered). The vector is treated as a categorical variable whose levels will be reordered. If `x` is not a factor, its unique values will be used as the implicit levels.
`X`	a vector of the same length as `x`, whose subset of values for each unique level of `x` determines the eventual order of that level.
`FUN`	a `function` whose first argument is a vector and returns a scalar, to be applied to each subset of `X` determined by the levels of `x`.
`...`	optional: extra arguments supplied to `FUN`
`order`	logical, whether return value will be an ordered factor rather than a factor.
`decreasing`	logical, whether the levels will be ordered in increasing or decreasing order.

Details

This, as `relevel()`, is a special case of simply calling `factor(x, levels = levels(x)[....])`.

Value

A factor or an ordered factor (depending on the value of `order`), with the order of the levels determined by `FUN` applied to `X` grouped by `x`. By default, the levels are ordered such that the values returned by `FUN` are in increasing order. Empty levels will be dropped.

Additionally, the values of `FUN` applied to the subsets of `X` (in the original order of the levels of `x`) is returned as the `"scores"` attribute.

Author(s)

Deepayan Sarkar deepayan.sarkar@r-project.org

See Also

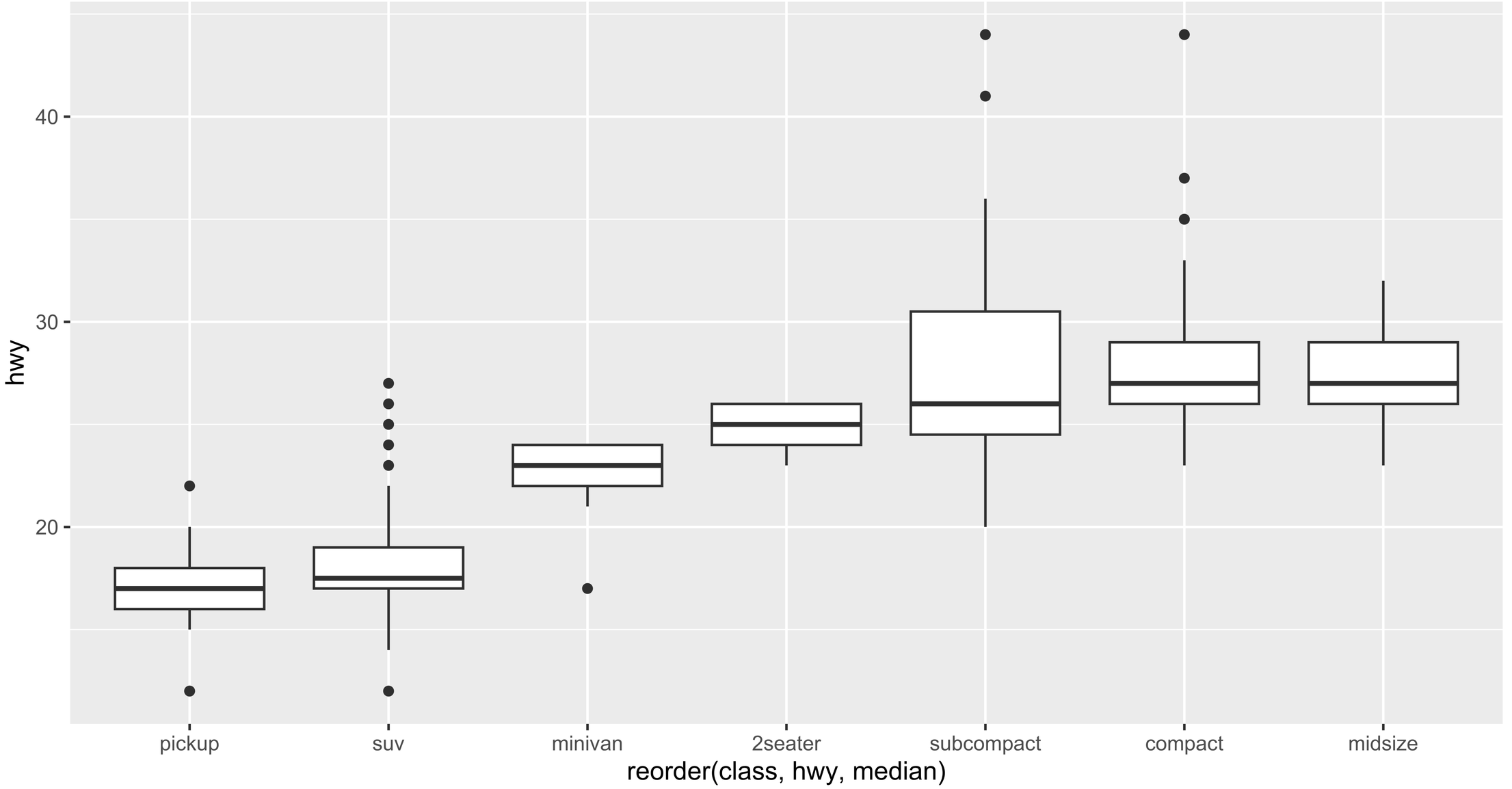
`reorder.dendrogram`, `levels`, `relevel`.

Examples

Run examples

```
require(graphics)

bymedian <- with(InsectSprays, reorder(spray, count, median))
boxplot(count ~ bymedian, data = InsectSprays,
        xlab = "Type of spray", ylab = "Insect count",
        main = "InsectSprays data", varwidth = TRUE,
```




```
ggplot( mpg, aes(x = reorder(class, hwy, median), y = hwy) ) +
  geom_boxplot()
```